Chapter 1

Course Overview

- Columns, rows, matrices, $\mathbb{R}^n$, $\mathbb{R}^{m\times n}$ are reserved for representational uses.
- These represent respectively,

  vectors, adjoint vectors, linear transformations, a real vector space X of dimensions n, $\mathsf{B}(\mathsf{X},\mathsf{Y})$ between X and Y.
- As seen above, in this course, it is almost exclusively the case of a vector space over the real field $\mathbb{R}$.
  1. It is necessary to understand the geometric point of view for $\mathbb{R}^{m\times n}$, which might be obscured by using $\mathbb{C}$ field.
  2. Linear algebra over $\mathbb{R}$ is in general more complicated to study than that over $\mathbb{C}$, and once we are done with $\mathbb{R}$ field, the corresponding material with $\mathbb{C}$ is usually immediate, but not in the other way around.
  3. We know linear algebra over finite field $\mathbb{Z}_p$ is useful in some mathematical areas. We will not cover them unfortunately.

- A vector space X can be of various kinds, that of colors, tastes, quantumn states, polynomials, $\cdots$, where the scalar multiple and addition make senses.

  For our expositional purpose, we mostly treat X as a vector space of

  $$\{\text{Locations with respect to a fixed origin}\} \quad \text{in } n\text{-dimensional physical spaces.}$$

  1. $n = 3$ case will be most intuitive for this.
  2. A location with a fixed origin can be scalar multipled, and two locations with a fixed orgin can be added by the parallelogram law.
  3. Once we attach the coordinate basis to the set of locations, a location can be specified by its coordinates, listed in a column.

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Orthogonal Matrices

- We will call a matrix $O \in \mathbb{R}^{m\times n}$ orthogonal if columns of O are orthonormal.
- We give a definition that makes sense even when O is a rectangular matrix. If O is orthogonal, it is necessarily $m \geq n$.
- For a square orthogonal matrix $Q \in \mathbb{R}^{n\times n}$ consisting of n orthonormal columns $q_1, q_2, \cdots, q_n$, it is obvious that

$$\begin{pmatrix} — & q_1^T & — \\ — & q_2^T & — \\ & \vdots & \\ — & q_n^T & — \end{pmatrix} \begin{pmatrix} | & | & \cdots & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & \cdots & | \end{pmatrix} = \mathrm{I}.$$

  In other words, Q^T is the inverse matrix of Q.

New representation after changing basis

Let X be the n-dimensional space of locations with fixed origin.

We will refer to the current coordinate basis

$$e_1, e_2, \cdots, e_n, \quad e_i = (0, 0, \cdots, 0, 1, 0, \cdots, 0)^T \quad 1 \text{ on } i\text{-th position}$$

as *native basis*, while some other basis elements in X

$$f_1, f_2, \cdots, f_n$$

as *foreign basis*.

A location $x \in \mathsf{X}$ is currently represented by column

$$c = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

We want to know the new column whose entries represents the location x but in foreign basis

$$c' = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix}.$$

To do this, (since it will be linear), we look for the native-to-foreign dictionary matrix $\square$ such that

$$c' = \square c$$

In many occasions, we are given foreign-to-native dictionary instead, that are

$$f_1, f_2, \cdots, f_n \quad \text{represented in native basis.}$$

In other words, the listing of foreign basis in columns,

$$F = \begin{pmatrix} | & | & \cdots & | \\ f_1 & f_2 & \cdots & f_n \\ | & | & \cdots & | \end{pmatrix}$$

is telling us in native language to a foreign word c': $Fc' = x$ is in native language that is c.

$$Fc' = c.$$

Hence, to get c', the native-to-foregin dictionary matrix $\square = F^{-1}$.

In our numerial linear algebra course, the most important base are orthonormal ones. Thus,

$$c' = Q^T c$$

explains the procedure obtaining the new representation after changing basis that are (orthonormal) columns of Q.

Rectangular Matrices

- A matrix $A \in \mathbb{R}^{m\times n}$ represents a linear mapping A (denoted without having another symbol)
$$A : \mathsf{X} \to \mathsf{Y},$$
a member of $\mathsf{B}(\mathsf{X}, \mathsf{Y})$ from X to Y of dimensions respectively n and m.
- There will be two parts of theory on rectangular matrices.
  1. $x \mapsto y = Ax$, the question is: what does A do on x to result in y?
  2. From y to the solution x. We will completely specify the set of all least square solutions.

The first part

Let A be any matrix. $A \in \mathbb{R}^{m\times n}$. To understand what does A do, we first of all wish to characterize the followings.

- The kernel of A
$$\mathsf{Ker}\,(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}.$$
- The range of A
$$\mathsf{Ran}\,(A) = \{y \in \mathbb{R}^m \mid y = Ax \quad \text{for some } x \in \mathbb{R}^n\}.$$
Considering the matrix-column multiplication
$$\begin{pmatrix} | & | & | & \cdots & | \\ c_1 & c_2 & c_3 & \cdots & c_n \\ | & | & | & \cdots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1c_1 + x_2c_2 + \cdots x_nc_n$$
we notice that
$$\mathsf{Ran}\,(A) = \mathsf{span}\,\langle c_1, c_2, \cdots, c_n\rangle.$$
- The range of A^T, the transpose of A
$$\mathsf{Ran}\,(A^T) = \{x \in \mathbb{R}^n \mid x = A^Ty \quad \text{for some } y \in \mathbb{R}^m\}.$$
If we write
$$A = \begin{pmatrix} — & r_1^T & — \\ — & r_2^T & — \\ & \vdots & \\ — & r_m^T & — \end{pmatrix} \quad \text{then} \quad \mathsf{Ran}\,(A^T) = \mathsf{span}\,\langle r_1, r_2, \cdots, r_m\rangle.$$
- The kernel of A^T
$$\mathsf{Ker}\,(A^T) = \{y \in \mathbb{R}^m \mid A^Ty = 0\}.$$

One immediately sees that

$$\begin{aligned} x \in \mathsf{Ker}\,(A) \quad &\text{if and only if} \quad x \perp \mathsf{span}\,\langle r_1, r_2, \cdots, r_m \rangle. \\ y \in \mathsf{Ker}\,(A^T) \quad &\text{if and only if} \quad y \perp \mathsf{span}\,\langle c_1, c_2, \cdots, c_n \rangle. \end{aligned}$$

In other words,

$$\mathbb{R}^n = \mathsf{Ker}\,(A) \oplus \mathsf{Ran}\,(A^T), \quad \mathbb{R}^m = \mathsf{Ker}\,(A^T) \oplus \mathsf{Ran}\,(A)$$

and they are orthogonal direct sums.

Fundamental Theorem of Linear Algebra:

Theorem 1. *For any matrix $A \in \mathbb{R}^{m \times n}$,*

$$dim\, \mathsf{span}\,\langle c_1, c_2, \cdots, c_n \rangle = dim\, \mathsf{span}\,\langle r_1, r_2, \cdots, r_m \rangle.$$

The equality of the two numbers gives rise to the definition of the rank.

- The rank of A
$$\mathsf{rank}\,(A).$$
- $\mathsf{rank}\,(A)$, $\mathsf{Ker}\,(A)$, $\mathsf{Ran}\,(A)$, $\mathsf{Ker}\,(A^T)$, and $\mathsf{Ran}\,(A^T)$ highlights about what happens in $x \mapsto Ax$.
- These will be studied in this course by
  1. QR decomposition
  2. Singular Value Decomposition.

The second part

The rootfinding part, that is to specify the complete set of all least square solutions, will be clear once we achieve QR-decomposition and Singular Value Decomposition.

Square Matrices

- In principle, all the materials for rectangular matrices applies to square matrices, where the vector spaces X and Y happen to have same dimensions.
- Inverse A^{-1} if exists, the determinant of A, $\cdots$ are additionally discussed.
- The particular part for square matrices are when we identify the domain space and the range space, that is X. $A \in \mathbb{R}^{n\times n}$ represents

$$A : \mathsf{X} \to \mathsf{X}$$

  In fancy words, A is an endomorphism, $A \in \mathsf{End}(\mathsf{X})$.

  1. Considering X as space of locations, $y = Ax$ is the new location in X referenced by $x \in X$. Then we can speak of shape deformation from the referenced shape $G \subset \mathsf{X}$ by A.
  2. We can then speak of rotation, reflection, dilation, $\cdots$.
  3. Most important concept in $\mathsf{End}(\mathsf{X})$ is the invariance under A. We look for $\mathsf{M} \subset \mathsf{X}$ such that $A\mathsf{M}$ does not escape M.

Invariant vector subspace under A

- In the simplest setting, this is the eigenvalue problem.
- We will treat two cases:
  1. Diagonalization for a symmetric matrix A.
  2. Schur decomposition for any square matrix A.

Today, we examine the fact and its consequences that

- there exists a stably-working numerical algorithm implementing
- diagonalization of a real square symmetric matrix.

1. We would like to talk about why this is a nontrivial fact.
2. Believing the fact, we show we will be able to, not only define but also compute

   $$\mathsf{rank}\,,\mathsf{Ran}\,,\mathsf{Ker}\,,\cdots$$

   for any $m \times n$ matrix.

Theorem. *For a $n \times n$ real symmetric matrix A, there exists an orthogonal matrix $O \in \mathbb{R}^{n\times n}$ and a diagonal matrix $\Lambda \in \mathbb{R}^{n\times n}$ such that*

$$A = O\Lambda O^T.$$

Singular Value Decomposition

- Here, we give a proof of the existence of Singular Value Decomposition for any matrix $A \in \mathbb{R}^{m\times n}$.
- The point of presenting this proof is to see if we can build an algorithm for SVD.

Theorem. *For any matrix $A \in \mathbb{R}^{m\times n}$, there exists a matrix $O \in \mathbb{R}^{m\times m}$, $\Sigma \in \mathbb{R}^{m\times n}$, and $Q \in \mathbb{R}^{n\times n}$ such that*

1. *O is an orthogonal matrix,*
2. *Q is an orthogonal matrix,*
3. *Σ is such that if $k = \min\{m, n\}$, the top left maximal $k \times k$ block is diagonal with diagonal entries*
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$$
and all other entries of Σ are zero,
4. *$A = O\Sigma Q^T$.*

Proof. .

1. We may give a proof for the case $m \geq n$. If not, we achieve
$$A^T = O\Sigma Q^T \implies A = Q\Sigma^T O^T$$
which is in the form as stated. Assume now $m \geq n$.

2. We consider a $n \times n$ symmetric matrix A^TA. We first check that
$$A^TAx = 0 \iff Ax = 0.$$
This is because
$$A^TAx = 0 \implies x^TA^TAx = 0 \implies |Ax|^2 = 0 \implies Ax = 0.$$
Converse is easy.

3. A^TA is a real symmetric and nonnegative definite. Hence, we have
$$A^TA = Q\Lambda Q^T.$$
The diagonalization can be arranged so that diagonal entries of Λ are in descending order:
$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0.$$
We let $r \leq n$ be the number such that
$$\lambda_r > 0 \quad \text{and} \quad i > r \implies \sigma_i = 0. \quad \text{Set } r = 0 \text{ if all eigenvalues are zero.}$$
and let $I_r = \{1, 2, \cdots, r\}$. The set may be empty if $r = 0$.
In particular, if $\lambda_i = 0$ then
$$\lambda_i q_i = A^TAq_i = 0 \implies Aq_i = 0.$$

4. Now, define $\mathbb{R}^m$-vectors for $i \in I_r$:

$$\text{for } i \in I_r \quad o_i = \frac{Aq_i}{\sqrt{\lambda_i}}.$$

We check

$$\text{for } i,j \in I_r, \quad o_i^T o_j = \frac{q_i^T A^T A q_j}{\sqrt{\lambda_i \lambda_j}} = \frac{\lambda_j q_i^T q_j}{\sqrt{\lambda_i \lambda_j}} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

5. If $\{r+1, r+2, \cdots, m\}$ is nonempty, supplement o_{r+1}, o_{r+2}, $\cdots$, o_m so that

$$o_1, o_2, \cdots, o_m \quad \text{are orthonormal.}$$

6. Let O, Q, Σ be such that: O consists of columns $o_1, o_2, \cdots, o_m$, Q consists of columns $q_1, q_2, \cdots q_n$, and Σ is as stated in Theorem with the top left diagonals

$$\sigma_i = \sqrt{\lambda_i}, \quad i \in I_r, \quad \text{and define } \sigma_{r+1} = \sigma_{r+2} = \cdots = \sigma_n = 0.$$

Finally, we show that

$$Ax = O\Sigma Q^T x \quad \text{for every } x \in \mathbb{R}^n.$$

Since $q_1, q_2, \cdots, q_n$ forms a basis of $\mathbb{R}^n$, let us suppose

$$x = c_1 q_1 + c_2 q_2 + \cdots c_n q_n.$$

In other words, $c = Q^T x$. Then

$$\begin{aligned}
Ax &= \sum_{i=1}^{n} c_i A q_i \\
&= \sum_{i \in I_r} c_i A q_i && (Aq_i = 0 \text{ for } i > r) \\
&= \sum_{i \in I_r} (\sigma_i c_i) o_i && (\text{definition of } o_i) \\
&= \sum_{i=1}^{n} (\sigma_i c_i) o_i && (\sigma_i = 0 \text{ for } i > r) \\
&= \sum_{i=1}^{n} (\Sigma Q^T x)_i o_i \\
&= \sum_{i=1}^{m} (\Sigma Q^T x)_i o_i && (\text{The component } (\Sigma Q^T x)_i = 0 \text{ for } i > n) \\
&= O\Sigma Q^T x.
\end{aligned}$$

□

For a given any matrix $A \in \mathbb{R}^{m\times n}$, we are given a Singular Value Decomposition

$$A = O\Sigma Q^T.$$

Then we can compute

- $\mathsf{rank}\,(A) = r$, where r is the number of nonzero singular values. If singular values are in descending order,
$$\sigma_r > 0 \quad \text{and} \quad i > r \implies \sigma_i = 0.$$
- $\mathsf{Ker}\,(A) = \mathsf{span}\left\langle q_{r+1}, q_{r+2}, \cdots q_n \right\rangle$.
- $\mathsf{Ran}\,(A) = \mathsf{span}\left\langle o_1, o_2, \cdots o_r \right\rangle$.
- $\mathsf{Ran}\,(A^T) = \mathsf{span}\left\langle q_1, q_2, \cdots q_r \right\rangle$.
- $\mathsf{Ker}\,(A^T) = \mathsf{span}\left\langle o_{r+1}, o_{r+2}, \cdots, o_m \right\rangle$.

These computations contrast their definitions

- $\mathsf{rank}\,(A)$, the number of independent columns (= number of independent rows)
- $\mathsf{Ker}\,(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$.
- $\mathsf{Ran}\,(A) = \{y \in \mathbb{R}^m \mid y = Ax \quad \text{for some } x \in \mathbb{R}^n\}$.
- and those for A^T.

Examine the steps 1-6 of the proof of the Singular Value Decomposition. Will we be able to build an algorithm?

It will still be a long journey until we achieve an algorithm for the diagonalization of a real symmetric square matrix.

Chapter 2

Square Matrices 1 : Elementary Square Matrices

Today, we begin Square Matrices 1.

- In general,
$$A : \mathsf{X} \to \mathsf{Y} \quad \text{is represented by a rectangular matrix.}$$
- In this section, we consider a special case
$$A : \mathsf{X} \to \mathsf{X} \quad \text{is represnted by a square matrix.}$$
In a proper words, $A \in \mathsf{End}(\mathsf{X})$.
To be concrete, we mostly consider the $n = 3$ case. One can consider $A \in \mathbb{R}^{3\times 3}$ as a geometric linear deformation of a shape.
- The point of speaking of elementary square matrices is that we do not need to know many kinds. There are only 5 kinds of linear deformation, and a general square matrix is a finite compositions of the 5 kinds.

  We only need to know 5 kinds of elementary types which we will study in this section.

  1. dilation,
  2. rotation,
  3. reflection,
  4. shearing,
  5. projection.
- You will build codes implementing the (square matrix-column) multiplication, in case the matrix is elemenatry. These codes will be building blocks for the algorighms in this course.

Very elementray square matrices → elementray square matrices → general square matrices.

Very elementary square matrices

These are elementary ones, and even based on the coordinate axes.

1. Dilation

We consider matrices of the form

$$\text{the identity matrix} \quad I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and the diagonal matrix} \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

For the moment, let us consider cases $\lambda_1, \lambda_2, \lambda_3$ are strictly positive.

This describes dilation with respect to x, y, and z axis by scale factors of λ_1, λ_2, and λ_3.

2. Reflection with respect to $n-1$ dimensional hyperplane

In 3-dimensions, there are 3 very elementary reflections, that are with repect to yz-plane, zx-plane, and xy-plane:

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

3. Orthogonal projections onto k-dimensional planes

These forms very elementary projection matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \cdots$$

onto xy-plane, x-axis, yz-plane, y-axis, $\cdots$.

4. Shearing

In 3-dimensions, there are very elementary shearings of the form

$$\begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

and in general an upper triangular matrix

$$\begin{pmatrix} \lambda_1 & t & r \\ 0 & \lambda_2 & s \\ 0 & 0 & \lambda_3 \end{pmatrix}.$$

- Shearing is a deformation that takes place with respect to cascading pivot axes: x-axis $\rightarrow$ xy-plane $\rightarrow$ xyz-space $\rightarrow$ $\cdots$

5. Rotation of 2-dimensinal subspaces keeping others invariant

In 3-dimensions, there are 3 very elementary rotations

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \quad \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

Generalizations into $\mathbb{R}^n$ cases can also be considered:

Elementary square matrices

- We have seen 5 kinds that are based on coordinate axes.
- Combining these with basis changing technique, we come up with new 5 kinds:
  1. If U is a very elementary square matrix and
  2. $Q \in \mathbb{R}^{n\times n}$ is the relevant orthogonal matrix, then
  $$T = QUQ^T \quad \text{is an elemenatry square matrix}$$
  3. based on new axes specified by columns of Q.
- We will examine one-by-one though below.
- We do this in the order: projection, reflection, rotation, dilation, shearings.

Orthogonal projections onto k-dimensional planes

- Suppose an orthogonal matrix Q is given in such a way that
  1. $\mathsf{span}\left\langle q_1, q_2, \cdots, q_k \right\rangle$ is the k-dimensional plane to project on.
  2. $\mathsf{span}\left\langle q_{k+1}, q_{k+2}, \cdots, q_n \right\rangle$ is the complementary orthogonal plane.
- Then the corresponding projection matrix ($n = 3$, $k = 2$ case)
$$T = Q \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} Q^T.$$
- In fact, if we only specify k vectors q_1, q_2, $\cdots$ q_k,
$$Q \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} Q^T = \left\{ \begin{pmatrix} | & | \\ q_1 & q_2 \\ | & | \end{pmatrix} \begin{pmatrix} — & q_1^T & — \\ — & q_2^T & — \end{pmatrix} \right\} = \hat{Q}\hat{Q}^T, \quad \hat{Q} = \begin{pmatrix} | & | \\ q_1 & q_2 \\ | & | \end{pmatrix}$$

Reflection with respect to $n-1$ dimensional hyperplane

- Suppose an orthogonal matrix Q is given in such a way that
  1. q_1 is unit normal to a certain $(n-1)$-dimensional hyperplane, and
  2. $\mathsf{span}\left\langle q_2, q_3, \cdots q_n \right\rangle$ is the hyperplane.
- Then the corresponding reflection matrix ($n=3$ case)

$$T = Q \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T.$$

- In fact, if we only specify the normal vector q_1,

$$Q \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T = Q \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\} Q^T = QQ^T - 2q_1 q_1^T$$
$$= \mathrm{I} - 2q_1 q_1^T.$$

Rotation of 2-dimensinal subspaces keeping others invariant

- Suppose an orthogonal matrix Q is given in such a way that
  1. $\mathsf{span}\left\langle q_1, q_2 \right\rangle$ is the 2-dimensional plane to rotate counterclockwisely, and
  2. $\mathsf{span}\left\langle q_3, q_4, \cdots, q_n \right\rangle$ are to be kept invariant.
- Then the corresponding rotation matrix ($n=3$ case)

$$T = Q \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T.$$

- In fact, if we only specify q_1 and q_2,

$$Q \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T = Q \left\{ \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\} Q^T$$
$$= \left\{ \hat{Q} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \hat{Q}^T \right\} + \mathrm{I} - \hat{Q}\hat{Q}^T, \quad \hat{Q} = \begin{pmatrix} | & | \\ q_1 & q_2 \\ | & | \end{pmatrix}$$

Dilation

- Suppose an orthogonal matrix Q is given in such a way that each q_i is axis of dialation.
- Then the corresponding dialation matrix is

$$T = Q \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} Q^T$$

- Matrices made in this way is symmetric.
- Conversely, according to Linear Algebra theory, any symmetric matrix can be put in the above form.

Shearings

- Suppose an orthogonal matrix Q is given in such a way that the cascading axes are q_1-axis $\rightarrow$ q_1q_2-plane $\rightarrow$ $q_1q_2q_3$-space $\rightarrow$ $\cdots$
- Then the corresponding shearing matrix is

$$T = Q \begin{pmatrix} \lambda_1 & t & r \\ 0 & \lambda_2 & s \\ 0 & 0 & \lambda_3 \end{pmatrix} Q^T$$

- Matrices made in this way is characterized by a square matrix whose eigenvalues are all real.
- Conversely, according to Linear Algebra theory, any such a matrix can be put in the above form.

Home Work 1: Important for later algorithms

1. Implement functions by yourself the elementary projection, reflection, and rotations.
   - Input: x and q_1, q_2, $\cdots$ that are relevant.
   - Output: y

Remark 2.1. .

- Those converse statements for dilation and shearing are subject of eigenvalues, invariant subspaces, $\cdots$. This will be treated in the last part of the course.
- We can state the following fact for square matrices though in advance.
- Let U be a very elementary square matrix, and consider T that can be put into

$$T = QUQ^T, \quad \underline{Q \text{ is orthogonal}}.$$

- The converse statements go in the correspondence below:

T :	{symmetric matices}	$\subset$	{matrices with real eigenvalues}	$\subset$	{all matrices}
U :	{diagonal matrices}	$\subset$	{upper triangular matrices}	$\subset$	{schur form matrices}

Home Work 2:

1. Verify that elementary reflection and elementary rotation matrices are orthogonal.

Remark 2.2. .

- Elementary reflection and elementary rotation matrices are orthogonal. The former has determinant -1 while latter has determinant $+1$.
- Any orthogonal matrix with determinant $+1$ can be made by finite times compositions of very elementary rotations.
- Any orthogonal matrix with determinant -1 can be made by above followed by one very elementary reflection.

Chapter 3

Rectangular Matrices 1
QR decomposition

- We are equipped with the elementary reflection, rotation, and projection.
- Then, we can do much stronger achievement, that is the QR decomposition.
- We very quickly achieve this, from one perspective of QR among many interpretations: we monitor $\mathsf{Ran}\,(A^T)$ while $\mathsf{Ran}\,(A)$ incidently follows at the end.
- After this quick achievement, we review material around QR from the begining again,
  1. other perspective where we monitor $\mathsf{Ran}\,(A)$ while $\mathsf{Ran}\,(A^T)$ incidently follows,
  2. meaning of them, numerical perspective too, and so on.

We recall matrix manipulation we have seen in Gauss Elimination.

- We say, A, A', A'', $\cdots$ are *Row equivalent* if

$$\mathsf{Ran}\,(A^T) = \mathsf{Ran}\,(A'^T) = \mathsf{Ran}\,(A''^T) = \cdots$$

- The objective of Gauss Elimination is to come up with
  1. simpler and simpler matrices but row equivalent.
     In most of textbook examples, this is understood as a job for human-students.
  2. Typically one turns A into R-form (defined below), further to Row Echelon Form, and finally to Reduced Row Echelon Form.

     Definition. *A matrix $A \in \mathbb{R}^{m\times n}$ is in R-form if*

$$A_{ij} = 0 \quad \textit{for } i > j.$$

- Let us for later purpose consider following: A, A', A'', $\cdots$ are *Column equivalent* if

$$\mathsf{Ran}\,(A) = \mathsf{Ran}\,(A') = \mathsf{Ran}\,(A'') = \cdots$$

Now, we give a statement of QR decomposition in the most general form.

Theorem 1. *For any $A \in \mathbb{R}^{m\times n}$, there exist $Q \in \mathbb{R}^{m\times m}$ and $R \in \mathbb{R}^{m\times n}$ such that*

1. *Q is orthogonal.*
2. *R is in Row Echelon Form.*
3. *$A = QR$.*

That we achieve Row Echelon Form includes that in the interemediate steps we identify what entries are pivots in R.

Apparent consequences of Theorem 1 in exact arithmetics

- Suppose we achieved

$$A = QR, \quad Q \text{ orthogonal, and } R \text{ REF.}$$

- This is compared to another REF produce by Gauss Elimination, where we write

$$A = PLU, \quad U \text{ is REF, } L \text{ is lower triangular, } P \text{ is a permutation.}$$

- We can read off immediately:
  1. $\mathsf{rank}\,(A)$ is the number of pivots in REF.
  2. $\mathsf{Ran}\,(A) = \mathsf{span}\left\langle \text{pivot columns of } Q \right\rangle$
  3. $\mathsf{Ran}\,(A^T) = \mathsf{span}\left\langle \rho_1, \rho_2, \cdots, \rho_r \right\rangle$ of nonzero rows of R. ρ_i is the i-th row of the REF R.
- Unlike the SVD, situation for the range space Y is certainly better than that of the domain space X.
  1. The pivot columns $q_{i(1)}, q_{i(2)}, \cdots q_{i(r)}$ are orthonormal. The complementary columns of Q spans $\mathsf{Ker}\,(A^T)$.
  2. The pivot rows $\rho_1, \rho_2, \cdots \rho_3$ are as in REF. To characterize $\mathsf{Ker}\,(A) = \mathsf{Ker}\,(R)$, we have to collect all solutions of
$$Rx = 0$$
- To determine a certain column is whether a pivot column or not is numerically not as perfect as we want. We can say **Revealing strength** in machine arithmetics, not in exact arithmetics, for $\mathsf{rank}\,(A)$, $\mathsf{Ran}\,(A)$, $\mathsf{Ran}\,(A^T)$ is as good as that of PLU decomposition.
- This is not as good as we want to achieve, but can be practical in many situations.

Very quick illustration for the algorithm for Theorem 1

- Either using the elementary rotations
- or using the elementary reflections

by pictures.

Algorithm summary by reflections:

Let $A \in \mathbb{R}^{m\times n}$.

- We compute Q_1, Q_2, $\cdots$, Q_k in $\mathbb{R}^{m\times m}$ so that

$$Q_k Q_{k-1} Q_{k-2} \cdots Q_1 A = R.$$

  ($Q\tilde{Q}$ is orthogonal if Q and $\tilde{Q}$ are orthogonal)

- Denoting

$$Q_1^T Q_2^T \cdots Q_k^T = Q, \quad A = QR.$$

  In fact, the elementary reflection matrix is symmetric:

$$Q = Q_1^T Q_2^T \cdots Q_k^T = Q_1 Q_2 \cdots Q_k.$$

Each Q_j examines β-th column c of $A_{j-1} = Q_{j-1}Q_{j-2}\cdots Q_1 A$, by the target index p:

If c is $\mathbb{R}^m$-column, let $c = c' + c''$,

$$c' = (c_1, c_2, \cdots, c_{p-1}, 0, 0, \cdots, 0)^T, \quad c'' = (0, 0, \cdots, 0, c_p, c_{p+1}, \cdots, c_m)^T.$$

Also, $\hat{c}'' = (c_p, c_{p+1}, \cdots, c_m)^T$.

1. Every prior column d of A_{j-1} are assumed: $d = d' + d''$ with $d'' = 0$.
2. Then $Q_j \in \mathbb{R}^{m\times m}$ is in the form

$$\left(\begin{array}{c|c} \mathrm{I}' & 0 \\ \hline 0 & Q_j'' \end{array}\right)$$

   (a) Multiplication $Q_j A_{j-1}$ leaves first $p-1$ rows of A_{j-1} invariant.
   (b) The other rows of A_{j-1} from p-th row are affected.
   (c) Note prior columns of A_{j-1} are kept invariant.

3. Computing Q_j'':

$$Q_j'' = \mathrm{I}'' - 2\hat{q}''\hat{q}''^T, \quad \hat{q}'' = \text{length normalized } \hat{\nu}''.$$

   $\hat{\nu}''$ is taken from one of

$$\hat{c}'' - |\hat{c}''|\,\hat{e}_1'' \quad \text{or} \quad \hat{c}'' + |\hat{c}''|\,\hat{e}_1''.$$

   If we want to choose the one with bigger length, it can be done in a handy way:

   (a) Since the two candidates differ only by the $\hat{e}_1''$ component,
   (b) The lengthier is the one with bigger (in absolute value) $\hat{e}_1''$ component,
   (c) We merely choose the sign aggravating the result. The copysign function works perfect here: (Double primes are omitted)

```
\hat\nu = \hat{c}
\hat\nu[1] += copysign(|\hat{c}|,\hat{c}[1])
```

If we take the simpler algorithm omitting the step deciding whether a certain column is a pivot column or not, we achieve mere R-form:

Theorem 2. *For any $A \in \mathbb{R}^{m\times n}$, there exist $Q \in \mathbb{R}^{m\times m}$ and $R \in \mathbb{R}^{m\times n}$ such that*

1. *Q is orthogonal.*
2. *R is in R-Form.*
3. *$A = QR$.*

If R is merely in R-form, not in REF, in general we cannot read off $\mathsf{Ran}\,(A)$ and $\mathsf{Ran}\,(A^T)$ out of

$$A = QR.$$

- Example:

$$A = \begin{pmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} | & | & | \\ q_1 & q_1 & q_1+q_2+q_3 \\ | & | & | \end{pmatrix}$$

  $\mathsf{rank}\,(A) = 2$, but how to compose $\mathsf{Ran}\,(A^T)$ and $\mathsf{Ran}\,(A)$ of 2-dimenions?

- If so, then for what purposes do we do this decomposition in Theorem 2?
- We achieve much less, in general, by Theorem 2. The objective of doing this will be specified when we review another perspective of QR decomposition: $\mathsf{Ran}\,(A)$-monitoring one.

Home Work :

1. Implement algorithms for Theorem 1 (with pivot identification) and Theorem 2 (without pivot identification)
2. Input : any matrix $A \in \mathbb{R}^{m\times n}$
3. Output : (Q, R), where $Q \in \mathbb{R}^{m\times m}$ orthogonal, $R \in \mathbb{R}^{m\times n}$ either REF or R-form.

Before we go further, we set the following result as our finest QR decomposition. This is called the QR decomposition + Column pivoting. (qrcp)

Theorem 3. *For any $A \in \mathbb{R}^{m\times n}$, there exist $O \in \mathbb{R}^{m\times m}$, $R \in \mathbb{R}^{m\times n}$, $Q \in \mathbb{R}^{n\times n}$ such that*

1. *O is orthogonal.*
2. *Q is orthogonal, in particular it is a permutation.*
3. *R is in Row Echelon Form.*
4. *$A = ORQ^T$.*

- We have said that **Revealing strength** in machine arithmetics for Theorem 1 may be practical.
- **Revealing strength** in machine arithmetics for Theorem 3 is as accurate as we want to achieve.

Home Work :

1. Implement algorithms for Theorem 3 with the column pivoting.

Remark 3.1. Comparing SVD ($A = O\Sigma Q^T$) to the decomposition in Theorem 3,

1. R is not as simple as the diagonal Σ in SVD while
2. Q is simpler here, that is just a permutation.

Review on QR decomposition

QR decomposition is the place where we can learn all the important concepts arising in numerical algorithms and their analysis.

More relevantly as a pedagogical purposes, we will visit one-by-one those concerns, and we will learn so many useful concepts.

Review 1. Same problem, Different perspective, Different algorithm

Of course, it is natural that there are many ways of resolving one same problem. QR decomposition example illustrates how choosing one particular way affects the later studies.

We now introduce one objective seemingly quite different from the one we had before:

- We turn $A \in \mathbb{R}^{m\times n}$ into A', A'', $\cdots$ that are column equivalent to each other.
- In particular we wish to have a property

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix}, \quad A' = \begin{pmatrix} | & | & \cdots & | \\ a'_1 & a'_2 & \cdots & a'_n \\ | & | & \cdots & | \end{pmatrix}$$

  so that

$$\begin{aligned} \mathsf{span}\left\langle a_1 \right\rangle &= \mathsf{span}\left\langle a'_1 \right\rangle \\ \mathsf{span}\left\langle a_1, a_2 \right\rangle &= \mathsf{span}\left\langle a'_1, a'_2 \right\rangle \\ \mathsf{span}\left\langle a_1, a_2, a_3 \right\rangle &= \mathsf{span}\left\langle a'_1, a'_2, a'_3 \right\rangle \\ &\vdots \\ \mathsf{span}\left\langle a_1, a_2, a_3, \cdots, a_n \right\rangle &= \mathsf{span}\left\langle a'_1, a'_2, a'_3, \cdots, a'_n \right\rangle. \end{aligned}$$

  If this is the case, and if all a'_j are orthonormal, this is to achieve the Gram-Schmidt Orthogonalization for a given family

$$a_1, a_2, \cdots, a_n.$$

- Since the family might not be independent, we expect less:

$$\begin{aligned} \mathsf{span}\left\langle a_1 \right\rangle &\subseteq \mathsf{span}\left\langle a'_1 \right\rangle \\ \mathsf{span}\left\langle a_1, a_2 \right\rangle &\subseteq \mathsf{span}\left\langle a'_1, a'_2 \right\rangle \\ \mathsf{span}\left\langle a_1, a_2, a_3 \right\rangle &\subseteq \mathsf{span}\left\langle a'_1, a'_2, a'_3 \right\rangle \\ &\vdots \\ \mathsf{span}\left\langle a_1, a_2, a_3, \cdots, a_n \right\rangle &\subseteq \mathsf{span}\left\langle a'_1, a'_2, a'_3, \cdots, a'_n \right\rangle. \end{aligned}$$

  This amounts to say that a_j is a linear combination of a'_1, a'_2, $\cdots$, a'_j, and thus

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ a'_1 & a'_2 & \cdots & a'_n \\ | & | & \cdots & | \end{pmatrix} R = A'R, \quad R \text{ is in R-form.}$$

- Note that, in this case it is natural to have: A' is $m \times n$, and R is $n \times n$.
  (Most relevant cases are when $m \geq n$.)
- This objective and our previous one has some in common.
- We will soon discuss that from numerical perspective, this way of achieving QR is inferior choice in many senses.

Review 2. QR **decomposition is an ill-conditioned problem.**

Testing error and Solution error are different kinds

It is easy to illustrate that the procedure $A \mapsto (Q, R)$ such that $QR = A$ is ill-conditioned.

By picture:

1. For an ill-conditioned problem, one cannot expect to build an algorithm with a small solution error.
2. However, this does not mean that one cannot build an algorithm with a small testing error. We actually can.

Let us be precise in saying that. Let us define the problem $\mathbf{P}_{QR}$**:**

1. Let the map

$$F : \{Q \in \mathbb{R}^{m\times m} \mid Q \text{ is orthogonal}\} \times \{R \in \mathbb{R}^{m\times n} \mid R \text{ is in R-form}\} \quad \to \quad \mathbb{R}^{m\times n}$$

$$F(Q, R) = QR,$$

2. We let $\mathsf{X} = \mathsf{X}_1 \times \mathsf{X}_2$, $\mathsf{Y} = \mathbb{R}^{m\times n}$

$$\mathsf{X}_1 = \{Q \in \mathbb{R}^{m\times m} \mid Q \text{ is orthogonal}\}, \quad \mathsf{X}_2 = \{R \in \mathbb{R}^{m\times n} \mid R \text{ is in R-form}\}.$$

3. For a given $A \in \mathsf{Y}$, find a solution of $F(Q, R) = A$.

The second item amounts to specify objectives:

2-(a) for $k = \min\{m, n\}$

$$\begin{aligned}
\mathsf{span}\left\langle a_1 \right\rangle &\subseteq \mathsf{span}\left\langle q_1 \right\rangle \\
\mathsf{span}\left\langle a_1, a_2 \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2 \right\rangle \\
\mathsf{span}\left\langle a_1, a_2, a_3 \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2, q_3 \right\rangle \\
&\vdots \\
\mathsf{span}\left\langle a_1, a_2, a_3, \cdots, a_k \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2, q_3, \cdots, q_k \right\rangle.
\end{aligned}$$

2-(b) q_1, q_2, $\cdots$, q_m are orthonormal.

We expect to be able to build an algorithm so that :

1. For given any $A \in \mathbb{R}^{m\times n}$, the algorithm H produces
$$H(A) = (\bar{Q}, \bar{R}).$$
2. The algorithm is consistent in the sense
   (a) $(\bar{Q}, \bar{R}) \in \mathsf{X}$. That is to say, $\bar{Q}$ is orthogonal and $\bar{R}$ is in R-form.
   (b) $\|A - \bar{Q}\bar{R}\| \leq \epsilon$

   for an appropriate $\epsilon > 0$. (The matrix norm will be discussed in the next chapter.)

However, this does not say that $(\bar{Q}, \bar{R})$ is close to any of true decomposition (Q, R) of A.

Review 3. The way of the 2nd approach can numerically fail to be consistent.

We consider what can happen if we proceed with the 2nd approach in cases

1. without pivot identification.
2. with pivot identification.
3. with column pivoting.

Up to the Elementary square matrices and QR decomposition, we have done the PART I.

To be able to perform what have been explained in the precise quantitative calculations, it leads to the necessity of studying concepts in the PART II.

1. Floating point numbers.
2. Matrix norm
3. Precision
4. Testing error estimates
5. Solution error estimates

   ⋮

We take the several QR decomp. algorithms as our examples to perform quantitative studies.

Chapter 4

Floating point numbers Machine arithmetics

$$\begin{aligned} \text{The model:} \quad & \big(\mathbb{R}, \star\big), && \star \in \{+, -, \times, \div\} \\ \text{Machine:} \quad & \big(\mathbb{F}, \star_{semi}\big), && \star_{semi} \in \{+_{semi}, -_{semi}, \times_{semi}, \div_{semi}\}. \end{aligned}$$

- It is important to recognize that what takes place in computer is semiconductor operation on discrete signals.
- More specifically, what are operated are floating point numbers in discrete and finite set
$$\mathbb{F} \subset \mathbb{R}.$$
- Action of $+, -, \times, \div$ on $\mathbb{R}$ are replaced by semiconductor operations on $\mathbb{F}$.

The set $\mathbb{F}$

- A member of $\mathbb{F}$ is called a floating point number.
- To tell what is the floating point number, we specify what is the scientific notation for a certain real number, first in Decimal representation.

Scientific notation in Decimal representation

- A real number $a \in \mathbb{R}$ for scientific computation is conventionally denoted in the form for examples
$$a = 6.7022 \times 10^8, \quad b = -1.200 \times 10^{-5}$$
- This notation reveals for a:
  1. The number of significant figures: 5, that are $6, 7, 0, 2, 2$.
  2. The number scale: 10^8.
  3. This tells that we are confident about the number a
$$[6.70215 \times 10^8, 6.0225 \times 10^8)$$

Of course, scientific notation in Binary representation are of our interests.

Floating point number

- The set $\mathbb{F}$ consists of numbers
(for the case 64 bits IEEE double precision)
$$(-1)^s \times 1.b \times 2^m,$$
  1. m is an integer in $[-1022, 1023]$,
  2. the printed digits b corresponds to an integer in $(0, 2^{52} - 1]$

  union two more elements: a sign bit followed by 63 zero bits, denoted by ± 0.
- More specific information on IEEE standard:
  1. Signals are operatated in a unit consiting of 64 bits.
  [S][EEEEEEEEEEE][FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF]
  2. 1 sign bit s
  3. 11 bits for nonnegative integer exponent $e \in [0, 2^{11} - 1] = [0, 2047]$.
  4. 52 bits for nonnegative integer $b \in (0, 2^{52} - 1]$
$$f = (-1)^s \times 1.b \times 2^{e-1023}.$$

Semiconductor operations

- We have $\star \in \{+, -, \times, \div\}$ on $\mathbb{R}$:

$$\star : \quad \mathbb{R} \times \mathbb{R} \quad \to \quad \mathbb{R}.$$

- Machine implements the semiconductor arithmetic operation $\star_{semi}$ on $\mathbb{F}$:

$$\star_{semi} : \quad \mathbb{F} \times \mathbb{F} \quad \to \quad \mathbb{F}, \quad \text{For } b, c \in \mathbb{F} \quad b \star_{semi} c \in \mathbb{F}.$$

The constant u

We depict the gap between $\mathbb{R}$ and $\mathbb{F}$, and between $\star$ and $\star_{semi}$ by abstrating a contant u.

Processing an expression

Processing an expression on floating point numbers in this course is explained by a map

$$\mathsf{fl} : expr \quad \mapsto \quad ans \in \mathbb{F} \cup \{\text{overflow}, \text{underflow}, \text{not a number}\}$$

$$\mathsf{fl}(x) = \begin{cases} \text{overflow} & \text{if } |x| \geq M \\ \text{underflow} & \text{if } |x| \leq \delta \\ \text{not a number} & \text{if } x \text{ is not defined} \\ b & \text{otherwise} \end{cases}$$

- Typically b is the closest floating number by rounding off.
- See more detail of IEEE 754.

We require a machine design the following consistency with respect to a constant u:

- There exists a bound $u > 0$ so that the following is true:

  (B1) For all $\delta \leq |x| \leq M$, there exists ϵ with $|\epsilon| \leq u$ such that

$$\mathsf{fl}(x) = x(1 + \epsilon).$$

  (B2) For all $b, c \in \mathbb{F}$, and $\star_{semi}$, there exists ϵ with $|\epsilon| \leq u$ such that

$$b \star_{semi} c = (b \star c)\ (1 + \epsilon).$$

Unless output falls into the exceptional cases,

- A number with 52, 53 significant figures in binary number system, can be accurately stored: with relative error bound scale

$$2^{-52} \simeq 2.22 \times 10^{-16}.$$

- For our usual computer, it is safe to consider

$$u = 4 \times 10^{-16}.$$

- The constant u abstracts the pseudo-exactness for the machine storing a value and performing arithmetics. It tells that if we perform arithmetics of numbers whose relative error bound are not finer than u, $\sim 10^{-16}$

  1. There will be no loss of significant figures attributed to machine arithmetics.
  2. Even if there are loss, we cannot blame the machine arithmetics.

Chapter 5

Rectangular Matrices 2
Matrix norms

We adopt notions for vector arithmetics.

Vector length

In our course, we almost exclusively use the Eucledan 2-lentgh for a vector $a \in \mathbb{R}^m$:

$$|a| = \big(a_1^2 + a_2^2 + \cdots + a_m^2\big)^{\frac{1}{2}}.$$

Frobenius norm of a matrix

Let $A \in \mathbb{R}^{m\times n}$. It is one particular way of listing in total mn real numbers in rows and columns.

Therefore, inherited from $\mathbb{R}^{mn}$, we let the inner product and norm

$$\big\langle A, B\big\rangle_F := \sum_{i=1}^{m}\sum_{\alpha=1}^{n} A_{i\alpha}B_{i\alpha}, \quad \|A\|_F := \big(\sum_{i,\alpha}(A_{i\alpha})^2\big)^{\frac{1}{2}}.$$

They are called Frobenius inner product and Frobenius norm.

Another norm of a matrix

Let

$$A : \mathbb{R}^n \to \mathbb{R}^m.$$

We denote

$$\|A\| = \sup_{x\in\mathbb{R}^n,\ |x|=1} |Ax|,$$

the supremum of length $|Ax|$ among x with length 1.

The Frobenius and the matrix norms are defined in abstract ways. In fact, they are concrete numbers.

- The matrix norm is the largest singular value σ_1 of A,

  $$\|A\| = \sigma_1(A).$$

  Considering the SVD

  $$A = O\Sigma Q^T, \quad \text{or} \quad O^T A = \Sigma Q^T$$

  Multiplying O^T or Q^T does not change the length in Y space and X space.

  The maximizer is attained, at $x = q_1$, and the maximum is σ_1.

- The Frobenius norm is

  $$\|A\|_F = \left(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2\right)^{\frac{1}{2}}, \quad k = \min\{m, n\}.$$

  This is because of the following obvious observation.

  $$\left\langle A, B \right\rangle_F = \mathsf{tr}\left(A^T B\right) = \mathsf{tr}\left(AB^T\right).$$

  If so, using SVD,

  $$\begin{aligned} \|A\|_F^2 = \mathsf{tr}\left(A^T A\right) &= \mathsf{tr}\left(Q\Sigma^T O^T O \Sigma Q^T\right) \\ &= \mathsf{tr}\left(Q\Sigma^T \Sigma Q^T\right) = \mathsf{tr}\left(\Sigma^T \Sigma\right) = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2 \end{aligned}$$

  since trace is invariant under the similarity transformation.

We will use the following inequalities:

1. $$\sigma_1 \le \left(\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2\right)^{\frac{1}{2}}, \quad \text{i.e.,} \quad \|A\| \le \|A\|_F.$$

2. For any $x \in \mathbb{R}^n$,

   $$|Ax| \le \|A\|\ |x| \le \|A\|_F\ |x|.$$

   It suffices to check this for $x \neq 0$:

   $$|Ax| = \left|A\frac{x}{|x|}\right|\ |x| \le \|A\|\ |x|.$$

3. For any $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times \ell}$,

   $$\left|ABx\right| \le \|A\|\ |Bx| \le \|A\|\ \|B\|\ |x| \quad \rightarrow \quad \|AB\| = \sup_{x\in\mathbb{R}^\ell,\ |x|=1} |ABx| \le \|A\|\ \|B\|.$$

4. For any $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times \ell}$, it also holds

   $$\|AB\|_F \le \|A\|_F \|B\|_F.$$

   This is because of the following:

Let $C = AB$ and denotes

(a) columns of A by $a_1, a_2, \cdots, a_n$,
(b) columns of B by $b_1, b_2, \cdots, b_\ell$,
(c) columns of C by $c_1, c_2, \cdots, c_\ell$,

noticing that

$$c_i = Ab_i.$$

Hence

$$|c_i|^2 = |Ab_i|^2 \le \|A\|^2 |b_i|^2$$
$$\implies \quad \sum_{i=1}^{\ell} |c_i|^2 \le \sum_{i=1}^{\ell} \|A\|^2 |b_i|^2, \quad \text{in other words} \quad \|C\|_F^2 \le \|A\|^2 \|B\|_F^2 \le \|A\|_F^2 \|B\|_F^2.$$

5. Suppose $A \in \mathbb{R}^{m\times n}$ and $m \ge n = r$, where $r = \mathsf{rank}\,(A)$. In this case,

$$\text{for any } x \in \mathbb{R}^n \quad \sigma_n |x| \le |Ax| \le \sigma_1 |x|.$$

Chapter 6

Precision of a number

The purpose of a few chapters from now are to develop tools to dicern the two implements in the following example.

Suppose we are to solve a quadratic equation

$$ax^2 + bx + c = 0, \quad \text{assmue } a \neq 0, \text{ and } b^2 - 4ac \geq 0$$

and let

$$H(a, b, c) = \Big(\frac{-b + \sqrt{b^2 - 4ac}}{2a}, \ \frac{-b - \sqrt{b^2 - 4ac}}{2a} \Big)$$

Let us consider its two implementations:

$$\tilde{H}_1(a, b, c) = \Big(\mathsf{fl}\Big(\frac{-b + \sqrt{b^2 - 4ac}}{2a} \Big) \qquad , \mathsf{fl}\Big(\frac{-b - \sqrt{b^2 - 4ac}}{2a} \Big) \Big)$$

$$\tilde{H}_2(a, b, c) = \Big(\mathsf{fl}\Big(\frac{-b - \text{sign}\,(b)\sqrt{b^2 - 4ac}}{2a} \Big) \qquad , \mathsf{fl}\Big(\frac{2c}{-b - \text{sign}\,(b)\sqrt{b^2 - 4ac}} \Big) \Big)$$

Passing expressions to the compositioning map fl, we now take the possibility that two outcomes are different. Thus, we develop one step further tools to assess the quality of the outcome passing to fl.

For a scientific computation, a number $a \in \mathbb{R}$ is not treated by itself alone, but always should be treated with its confidence interval, for example

$$3256.83 \pm 0.02.$$

A scientific computation can be divided by a sequence of steps

$$a_0 \quad \mapsto \quad a_1 = H_1(a_0) \quad \mapsto \quad a_2 = H_2(a_1) \quad \mapsto \quad \cdots \quad \mapsto \quad b = H_k(a_{k-1})$$

to achieve an objective. For a step solver H_j, it admits upstream values, perform some computations, and passes some results to downstream.

In principle, we wish to preserve a certain *precision measure*, such as number of significant figures, from the first prepared data all the way down to the final output. This in fact demands the preservation of that precision measure at every steps.

Now we specify what does it mean by the word precision.

Notations

In Euclidean space $\mathbb{R}^m$, the closed ball centered at $c \in \mathbb{R}^m$ with radius $r > 0$ is denoted by

$$\bar{B}(c, r).$$

We also consider a constant multiple of a ball,

$$\lambda\bar{B}(c, r) = \{\lambda x \mid x \in \bar{B}(c, r)\}.$$

sample and sample ball

Recall the expression

$$3256.83 \pm 0.02.$$

We will call the 3256.83 a nominal value, and the interval (1-d ball)

$$[3256.83 - 0.02,\ 3256.83 + 0.02]$$

its *sample space*. In this course, we restrict ourselves to sample space that is a ball. We let

$$S_a = \text{the set of elements } a \text{ is representing.}$$

and a choice of an element of S_a will be denoted by $s(a)$.

It is the sample ball of a nominal value a that decides a certain precision measure.

The relative error bound for a nonzero number

Precision by relative error bound

The fraction how large the error relatively to the total

$$w = \frac{|0.02|}{|3256.83|}$$

is called the relative error bound.

If a precision for a is given by a way specifying its relative error bound w, then it amounts to say in terms of the sample ball

$$s(a) \in |a|\bar{B}\Big(\frac{a}{|a|}, w\Big).$$

Precision by the number of significant figures

The number of significant figures for a nonzero a indicates the relative error bound.

$$\text{for} \quad a = -1.200 \times 10^{-5}, \quad \text{it has 4 significant figures.}$$

In terms of the sample ball, we can take

$$s(a) \in 10^{-5}\bar{B}\big(-1.200, \quad 0.5 \times 10^{-4+1}\big).$$

For the relative error bound,

$$w \leq \frac{0.5 \times 10^{-4+1} \times 10^{-5}}{1.200 \times 10^{-5}} = \frac{0.5 \times 10^{-4+1}}{1.200} \leq 0.5 \times 10^{-4+1}.$$

If a has k total significant figures in Decimal representation, then we can take a relative error bound such that

$$w \leq 0.5 \times 10^{-k+1}.$$

Representing a sample for given relative error bound

The relative error bound is utilized by representing a sample as an equality

$$s(a) = a\big(1 + w_1\big), \quad |w_1| \leq w. \quad \text{Or, equivalently} \quad s(a) = |a|\big(\text{sign}\,(a) + w_2\big), \quad |w_2| \leq w.$$

A simple data set $\big(\mathbb{A}, M, w\big)$

- To proceed our quantative calculations, in this course we consider a set $\mathbb{A}$ of numbers whose precision is specified both by the relative and absolute error in the following sense.
- For examples, we can collect elements whose number of siginicant figures are at most 6 ($w = 0.5 \times 10^{-6+1}$), and in a typical scale $M = 10^3$.

$$\begin{aligned} 4321.05 \\ -1328.78 \\ -246.895 \\ 9900.00 \\ 1.37152 \\ \vdots \end{aligned}$$

  We assume that we are not capable to resolve a digit beyond $10^3 \times 10^{-6+1} = 10^{-2}$.

In this course, we consider a prepared upstream data set $\mathbb{A}$ equipped with the following.

- We assume $\mathbb{A}$ has a single typical scale M, and $a \in \mathbb{A}$ is relatively written as

$$\mathring{a} = \frac{a}{M}.$$

  Assume $a \in \mathbb{A}$ is not too big relative to M.
- We let the sample ball radius for $\mathring{a}$ be w:

$$s(a) \in M\bar{B}(\mathring{a}, w).$$

- This fact will be utilized by writing a sample

$$s(a) = M(\mathring{a} + w_1), \quad |w_1| \le w$$

- For examples in the above, we may let

$$\begin{aligned} M = 10^3, \quad w = 0.5 \times 10^{-6+1}, \\ s(4321.05) = 10^3 \times \big(4.32105 + w_1\big), \qquad & |w_1| \le w \\ s(-1328.78) = 10^3 \times \big(-1.32878 + w_2\big), \qquad & |w_2| \le w \\ s(1.37152) = 10^3 \times \big(0.00137152 + w_3\big), \qquad & |w_3| \le w \\ \vdots \end{aligned}$$

Interpretation of the form of error in terms of significant figures

When we write

$$s(a) = M(\mathring{a} + w_1) \quad |w_1| \leq w, \quad \text{or equivalently} \quad s(a) \in M\bar{B}(\mathring{a}, w)$$

it represents a sample revealing the form of error. In terms of significant figures, we conclude the following:

1. Case $\mathring{a} = \frac{a}{M}$ is of typical scale $\sim 10^0$.
   - For this case, w plays the role of the relative error bound,
   - In the above example, the nominal a will be understood to carry the 6 significant figures, as same as interepreted by the relative error bound.
2. Case $\mathring{a}$ is of subscale $\sim 10^{-k}$ for some positive k.
   - As seen in
     $$s(1.37152) = 10^3 \times \big(0.00137152 + w_3\big)$$
     if $w_3 \simeq 10^{-6+1}$, the digits beyond that are meaningless. Then, only 1, 3, and 7 from left will be significant figures.
   - There are loss of k significant figures.
   - As an extream case, let us consider the sample ball of the nominal zero:
     $$s(0) = 10^3 \times \big(0 + w_4\big).$$
     The nominal zero may represent a sample whose length is less than $Mw = 0.005$, and we may say no siginificant figure exists in this case.

Of course, precision can be measured by many other various ways.

We find it convenient to use a simple data set $(\mathbb{A}, M, w)$ with a precision since the previous precision by relative error bound for a nonzero element can be treated as well by this framework: by taking a singletone set $\mathbb{A} = \{a\}$ and $M = |a|$.

Chapter 7

Condition and Consistency for a step H

We are ready to speak of condition and consistency of a certain (step) solver H.

Let H resolve a certain step

$$H : \mathbb{A} \to \mathbb{B},$$

where $\mathbb{A}$ is the upstream data set, and $\mathbb{B}$ is the downstream data set.

We also assume an approximate solver or an algorithm $\tilde{H}$ that implements H.

Roughly speaking,

1. A step H is well-conditioned if it preserves the *precision* of upstream data.

2. An approximate solver $\tilde{H}$ is consistent with respect to the precision of $\mathbb{A}$ if its approximate output $\tilde{b} = \tilde{H}(a)$ is materialized by one of the sample scenario of a under the true solver, that is $H(s(a))$.

We will use more specifically the terminology by narrowing the senses.

Condition of $H : \mathbb{A} \to \mathbb{B}$

Suppose $a \in \mathbb{A}$ and $H(a) = b \in \mathbb{B}$ are both nonzero.

Suppose a precision of a is given by the relative error bound w.

The condition of H at a is decided by the following ratio:

$$\kappa_{H,w,a} = \frac{w'}{w}, \quad w' = \inf\Big\{\rho \mid H\Big(|a|\bar{B}(\tfrac{a}{|a|}, w)\Big) \subset |b|\bar{B}(\tfrac{b}{|b|}, \rho)\Big\}.$$

Equivalently

$$w' = \sup_{s(a)} \frac{|H(s(a)) - b|}{|b|} = \sup_{\frac{|\tilde{a}-a|}{|a|} \le w} \frac{|H(\tilde{a}) - b|}{|b|}.$$

Then, the sample ball for the resultant value b will be:

$$s(b) \in |b|\bar{B}\big(\tfrac{b}{|b|}, w'\big), \quad w' = \kappa_{H,w,a} w.$$

i.e., the relative error bound w is multipled by the number $\kappa_{H,w,a}$.

We may want to have its limit value, depending only on H and a

$$\kappa_{H,a} = \limsup_{w \to 0} \kappa_{H,w,a}.$$

This number can be said to be the condition number of the step H at a.

If $\kappa_{H,w,a}$ are well-defined for all of $\mathbb{A}$, then we may as well define

$$\kappa_{H,w} = \sup_{a \in \mathbb{A}} \kappa_{H,w,a}.$$

One can do both, in one order or in the other order.

Difference from continuity

- The well-conditionedness asks whether precision is preserved. It is related to the continuity but the following example illustrates the difference from the continuity. Let

  $$H(x) = 1 + \sqrt{|x-1|} \quad \text{near } x = 1.$$

  See that $H(1) = 1$. For $s(x) \in B(1, w)$ with $s(x) \neq 1$, we compute the ratio

  $$\frac{|H(s(x)) - H(1)|/H(1)}{w} \le \frac{\sqrt{w}}{w} = \frac{1}{\sqrt{w}} =: \kappa_{H,w,x}, \quad \kappa_{H,w,x} \to \infty \quad \text{as } w \to 0.$$

- Near 1, the mapping H loses significant figures, although it is continuous at 1. The condition number becomes unbounded as $w \to 0$.

One may as well use the simple data set $(\mathbb{A}, M, w)$. We will introduce this in concrete examples.

Consistency of $\tilde{H}$ on $\mathbb{A}$.

Suppose $a \in \mathbb{A}$ is nonzero and $H(a) = b \in \mathbb{B}$.

Suppose a precision of a is given by the relative error bound w.

We say $\tilde{H}$ is consistent at a with respect to the precision w if

1. $\tilde{b} = \tilde{H}(a) \in \mathbb{B}$.
2. $\tilde{b} = H(s(a))$ for some $s(a) \in |a|\bar{B}\big(\frac{a}{|a|}, w\big)$.

Suppose a precision is given by fixing $\big(\mathbb{A}, M, w\big)$.

We say $\tilde{H}$ is consistent at $a \in \mathbb{A}$ with respect to $\big(\mathbb{A}, M, w\big)$.

1. $\tilde{b} = \tilde{H}(a) \in \mathbb{B}$.
2. $\tilde{b} = H(s(a))$ for some $s(a) \in M\bar{B}\big(\mathring{a}, w\big)$.

This is related to the testing error bound we discussed in the 1st semester. Suppose we are to construct a solver for an equation

$$F(x) = y, \quad \text{with inverse } F^{-1} = H.$$

We let H be the solver, input value $a = y$, and output value $b = x$. Assume an approximate solver $\tilde{H}$. The above consistency applied to this case is saying that

1. The approximate solution $\tilde{b} = \tilde{H}(a) \in \mathbb{B}$.
2. The testing error, measured in relative sense, $\frac{|F(\tilde{b})-a|}{|a|} \leq w$.

Remark 7.1. The standard terminology for this consistency is the *backward stability*. We refrain from using it at least in this course because:

1. The word *stability* is used to mean so many different things in numerical analysis. We find it right to use stability when a small perturbation of input merely leads to a small perturbation of output, namely the continuity. Here, it is used for totally different meaning.
2. To speak of *backward direction* we have to first specify what is the *forward direction*. We find it less confusing rather not to speak of any of direction.
3. Consistency is a notion concerning the relevance/irrelevance of an approximation to the true one. If irrelevant, i.e., inconsistent, then nothing is valuable out of it.

Chapter 8

Examples: a form of errors after floating point arithmetics

In this chapter, we compute the condition of the arithmetics of real numbers.

We consider

$$\star_{semi} : \mathbb{A}_1 \times \mathbb{A}_2 \to \mathbb{B}$$

To this purpose, we consider the simple data set specified with the precision

$$\big(\mathbb{A}_1, M_1, w\big), \quad \big(\mathbb{A}_2, M_2, w\big).$$

The precision by relative error bound can be considered by taking the singletone sets

$$\mathbb{A}_1 = \{a_1\}, \quad M_1 = |a_1|, \quad \mathbb{A}_2 = \{a_2\}, \quad M_2 = |a_2|.$$

Multiplication

- We consider the preparation of nominal a_1 and a_2 with

$$s(a_1) = M_1(\mathring{a}_1 + w_1), \quad s(a_2) = M_2(\mathring{a}_2 + w_2), \quad |w_1|, |w_2| \leq w.$$

- For the multiplication, we find $M_3 = M_1 M_2$ for the resultant typical scale appropriate.

  We consider $\left(\mathbb{B}, M_3, w'\right)$ and now we seek for the bound w' below.

- We first proceed with nonzero $a_1 \in \mathbb{A}_1$ and nonzero $a_2 \in \mathbb{A}_2$ as described in the above by singletone case.

  1. We consider the preparation of nominal a_1 and a_2 with

  $$s(a_1) = |a_1|(\mathring{a}_1 + w_1), \quad s(a_2) = |a_2|(\mathring{a}_2 + w_2), \quad \mathring{a}_1 = \text{sign}\,(a_1), \quad \mathring{a}_2 = \text{sign}\,(a_2).$$

  2. Then one sample scenario gives

  $$\begin{aligned} s(a_1) \times_{semi} s(a_2) &= s(a_1)s(a_2)(1+u) \text{we assume } u \leq w \text{ below} \\ &= |a_1||a_2|(\mathring{a}_1 + w_1)(\mathring{a_2} + w_2)(1+u) \\ &= |a_1||a_2|(\mathring{a}_1\mathring{a}_2 + w_1\mathring{a_2} + w_2\mathring{a_1} + w_1 w_2)(1+u) \\ &= |a_1||a_2|(\mathring{a}_1\mathring{a}_2 + \kappa w) \\ &= |a_1||a_2|(\mathring{b} + w'), \quad \mathring{b} = \text{sign}\,(a_1 a_2). \end{aligned}$$

  where $w' = \kappa w$ with κ not too big.

  3. We see $|\mathring{b}| = 1$ is not of sub-scale. Therefore, the condition in terms of the relative error bound for the multiplication is sufficiently good. There are no loss of significant figures.

- In case one of the nominal input is zero, then the form of error is

$$s(a_1)s(a_2) = M_3(0 + \kappa w),$$

  the result is the nominal zero with no significant figure, that may represents a number whose size $\simeq M_3 \kappa w$.

Division

- We consider the preparation of nominal a_1 and $a_2 \neq 0$ with

$$s(a_1) = M_1(\mathring{a}_1 + w_1), \quad s(a_2) = M_2(\mathring{a}_2 + w_2), \quad |w_1|, |w_2| \leq w.$$

- For the division, we find $M_3 = \frac{M_1}{M_2}$ for the resultant typical scale appropriate.

  We consider $\big(\mathbb{B}, M_3, w'\big)$ and now we seek for the bound w' below.

- We first proceed with nonzero $a_1 \in \mathbb{A}_1$ as described in the singletone case above.

  1. We consider the preparation of nominal a_1 and a_2 with

  $$s(a_1) = |a_1|(\mathring{a}_1 + w_1), \quad s(a_2) = |a_2|(\mathring{a}_2 + w_2), \quad \mathring{a}_1 = \text{sign}\,(a_1), \quad \mathring{a}_2 = \text{sign}\,(a_2).$$

  2. Then one sample scenario gives

$$\begin{aligned} s(a_1) \div_{semi} s(a_2) &= \frac{s(a_1)}{s(a_2)}(1+u) \\ &= \frac{|a_1|}{|a_2|}\Big(\frac{\mathring{a}_1 + w_1}{\mathring{a}_2 + w_2}\Big)(1+u) \\ &= \frac{|a_1|}{|a_2|}\Big(\frac{\mathring{a}_1}{\mathring{a}_2} + \kappa w\Big) \\ &= \frac{|a_1|}{|a_2|}\Big(\mathring{b} + w'\Big), \quad \mathring{b} = \text{sign}\,\frac{a_1}{a_2} \end{aligned}$$

  where $w' = \kappa w$ with κ not too big.

  3. We see $|\mathring{b}| = 1$ is not of sub-scale. Therefore, the condition with respect to the relative error bound for the division is sufficiently good: There are no loss of significant figures.

- In case the nominal input a_1 is zero, then the form of error is

$$s(a_1) \div_{semi} s(a_2) = M_3(0 + \kappa w)$$

  the result is the nominal zero with no significant figure, that may represents a number $\simeq M_3 \kappa w$.

Including the zero output case, there is nothing wrong in this examination, but one concern is that, if the typical scale for the denominator is of tiny scale and if one is not aware of this is so, then the absolute error scale

$$M_3 w' = \frac{M_1}{M_2} w'$$

can be very big than one's (wrong) expectation !

Addition and Subtraction

- We consider the preparation of nominal a_1 and a_2 with

$$s(a_1) = M_1(\mathring{a}_1 + w_1), \quad s(a_2) = M_2(\mathring{a}_2 + w_2), \quad |w_1|, |w_2| \leq w.$$

- For the addition and substraction, we find $M_3 = \sqrt{M_1^2 + M_2^2}$ for the resultant typical scale appropriate.

  We consider $\big(\mathbb{B}, M_3, w'\big)$ and now we seek for the bound w' below.

- Then one sample scenario gives

$$\begin{aligned} s(a_1) +_{semi} s(a_2) &= \big(s(a_1) + s(a_2)\big)(1+u) \\ &= \sqrt{M_1^2 + M_2^2}\Big(\frac{M_1\mathring{a}_1}{\sqrt{M_1^2 + M_2^2}} + \frac{M_2\mathring{a}_2}{\sqrt{M_1^2 + M_2^2}} + \frac{M_1 w_1 + M_2 w_2}{\sqrt{M_1^2 + M_2^2}}\Big)(1+u) \\ &= \sqrt{M_1^2 + M_2^2}\Big(\frac{M_1\mathring{a}_1}{\sqrt{M_1^2 + M_2^2}} + \frac{M_2\mathring{a}_2}{\sqrt{M_1^2 + M_2^2}} + \kappa w\Big) \\ &= \sqrt{M_1^2 + M_2^2}\big(\mathring{b} + w'\big). \end{aligned}$$

  where $w' = \kappa w$ with κ not too big.

- The nominal result $b = a_1 + a_2$ is written relatively in the typical scale

$$\mathring{b} = \frac{b}{\sqrt{M_1^2 + M_2^2}}.$$

  We see that downscaling may happen: we do not have a lower bound of for $|\mathring{b}|$.

  1. If the down scaling took place, then in terms of relative error bound, this loses significant figures. This is natural: in a subtraction with 4 significant figures

$$4321 - 4320 = 1$$

     and the outcome has only one significant figure, where the loss has been accompanied by down scaling of typical scale from 10^3 to 10^0.
  2. Only if $\mathring{b} \simeq 10^0$ or in higher scale, loss of significant figures does not take place.

The number of significant figures are preserved in real number arithmetics, provided that the nominal result is "vivid" relative to the resultant scale.

Now we recall the implementations of quadratic equation solver

$$\begin{aligned} \tilde{H}_1(a,b,c) &= \Big(\mathsf{fl}\Big(\frac{-b + \sqrt{b^2 - 4ac}}{2a}\Big) \qquad , \mathsf{fl}\Big(\frac{-b - \sqrt{b^2 - 4ac}}{2a}\Big)\Big) \\ \tilde{H}_2(a,b,c) &= \Big(\mathsf{fl}\Big(\frac{-b - \operatorname{sign}(b)\sqrt{b^2 - 4ac}}{2a}\Big) \qquad , \mathsf{fl}\Big(\frac{2c}{-b - \operatorname{sign}(b)\sqrt{b^2 - 4ac}}\Big)\Big) \end{aligned}$$

We worry the case $|ac| \ll |b|$. Now you can tell that the two are not same, and the second one will not lose the significant figures.

Chapter 9

Precision of vector and matrix

As vectors and Matrices are in normed vector spaces, the precision can be defined by sample balls in the same fashion:

Vector

We consider a set $\mathbb{A}$ of vectors in $\mathbb{R}^p$. Typically for a singletone set $\mathbb{A} = \{x\}$, we let

$$M = |x|, \quad \mathring{x} = \frac{x}{|x|}.$$

More generally, by writing $\big(\mathbb{A}, M, w\big)$ with $x \in \mathbb{A}$ we assume the sample

$$s(x) \in M\bar{B}(\mathring{x}, w).$$

Matrix

We consider a set $\mathbb{A}$ of of matrices in $\mathbb{R}^{m\times n}$. Typically for a singletone set $\mathbb{A} = \{A\}$, we let

$$M = \|A\|, \quad \mathring{A} = \frac{1}{\|A\|}A \qquad \text{or} \qquad M = \|A\|_F, \quad \mathring{A} = \frac{1}{\|A\|_F}A$$

More generally, by writing $\big(\mathbb{A}, M, w\big)$ with $A \in \mathbb{A}$ we assume the sample

$$s(A) \in M\bar{B}(\mathring{A}, w).$$

The form of an error

$$\begin{aligned} &\text{for } x \in \mathbb{R}^p && s(x) = M(\mathring{x} + w_1 v), \quad |w_1| \le w, \quad |v| \le 1. \\ &\text{for } A \in \mathbb{R}^{m\times n} && s(A) = M(\mathring{A} + w_2 E), \quad |w_1| \le w, \quad \|E\|_F \le 1. \end{aligned}$$

Remark 9.1. This definition is a precision as a whole. $|\mathring{x}| \simeq 1$ does not imply entrywisely i-th component $|\mathring{x}_i| \simeq 1$. Hence entrywise relative error bound might be worse than the relative error bound as a whole.

Chapter 10

Error propagation form on linear algebraic operations

In this chapter, we assume the simple data set specified with the precision

$$\big(\mathbb{A}_1, M_1, w\big), \quad \big(\mathbb{A}_2, M_2, w\big).$$

Pairing

- We consider the operation $(c, x) \mapsto c^T x \in \mathbb{R}$ for given c, x in an Euclidean space:

$$H : \mathbb{A}_1 \times \mathbb{A}_2 \to \mathbb{B}$$

- We consider the preparation of nominal $c, x \in \mathbb{R}^m$ with

$$s(c) = M_1(\mathring{c} + w_1 v_1), \quad s(x) = M_2(\mathring{x} + w_2 v_2), \quad |w_1|, |w_2| \le w, \quad |v_1|, |v_2| \le 1.$$

  Here, $w_1, w_2 \in \mathbb{R}$ and $v_1, v_2 \in \mathbb{R}^m$.

- For the multiplication, we find $M_3 = M_1 M_2$ for the resultant typical scale appropriate.

  We consider $\big(\mathbb{B}, M_3, w'\big)$ and now we seek for the bound w' below.

- We first proceed with nonzero $c \in \mathbb{A}_1$ and nonzero $x \in \mathbb{A}_2$.

  1. We consider the preparation of nominal a_1 and a_2 with

$$s(c) = |c|(\mathring{c} + w_1 v_1), \quad s(x) = |x|(\mathring{x} + w_2 v_2), \quad \mathring{c} = \frac{c}{|c|}, \quad \mathring{x} = \frac{x}{|x|}.$$

  2. Then one sample scenario via semiconductor arithmetics gives

$$\begin{aligned} s(c)^{T_{semi}} s(x) &= \sum_{i=1}^{m} s(c)_i s(x)_i (1 + u_i) \quad \text{we assume } u \le w \text{ below} \\ &= |c||x| \big\{ (\mathring{c} + w_1 v_1)^T (\mathring{x} + w_2 v_2) \big\}_i (1 + u_i) \\ &= |c||x| \big(\mathring{c}^T \mathring{x} + w_1 v_1^T \mathring{x} + w_2 \mathring{c}^T v_2 + w_1 w_2 v_1^T v_2 + w_3 \big) \\ &= |c||x| \big(\mathring{c}^T \mathring{x} + \kappa w \big) \\ &= |c||x| (\mathring{b} + w'), \quad \mathring{b} = \frac{c^T x}{|c||x|}. \end{aligned}$$

  where $w' = \kappa w$ with κ not too big.

3. We see that downscaling may happen for the relative nominal output $\mathring{b}$: we do not have a lower bound of for $|\mathring{b}|$.
   (a) If the down scaling took place, then in terms of relative error bound, this loses significant figures.
   (b) Only if $\mathring{b} \simeq 10^0$ or in higher scale, loss of significant figures does not take place.

- In case one of the nominal input is zero, then the form of error is

$$s(c)^T s(x) = M_3(0 + \kappa w),$$

the result is the nominal zero with no significant figure, that may represents a number whose size $\simeq M_3 \kappa w$.

Matrix-vector multiplication

- We consider the operation $(A, x) \mapsto Ax$ for given A, x:

$$H : \mathbb{A}_1 \times \mathbb{A}_2 \to \mathbb{B}$$

- We consider the preparation of nominal A, x

$$s(A) = M_1(\mathring{A} + w_1 E), \quad s(x) = M_2(\mathring{x} + w_2 v), \quad |w_1|, |w_2| \le w, \quad \|E\|, |v| \le 1.$$

  Here, $w_1, w_2 \in \mathbb{R}$ and $E \in \mathbb{R}^{m \times n}, v \in \mathbb{R}^n$.

- For the multiplication, we find $M_3 = M_1 M_2$ for the resultant typical scale appropriate.

  We consider $\big(\mathbb{B}, M_3, w'\big)$ and now we seek for the bound w' below.

- We first proceed with nonzero matrix $A \in \mathbb{A}_1$ and nonzero vector $x \in \mathbb{A}_2$.

  1. We consider the preparation of nominal A and x with

$$s(A) = \|A\|(\mathring{A} + w_1 E), \quad s(x) = |x|(\mathring{x} + w_2 v), \quad \mathring{A} = \frac{A}{\|A\|}, \quad \mathring{x} = \frac{x}{|x|}.$$

  2. Then one sample scenario via semiconductor arithmetics gives

$$\begin{aligned} s(A)s(x) &= \|A\||x|\big((\mathring{A} + w_1 E)(\mathring{x} + w_2 v) + w_3 v_2\big) \\ &= \|A\||x|\big(\mathring{A}\mathring{x} + w_1 E\mathring{x} + w_2 \mathring{A} v + w_1 w_2 E v + w_3 v_2\big) \\ &= \|A\||x|\big(\mathring{A}\mathring{x} + \kappa w v_3\big) \\ &= \|A\||x|(\mathring{b} + w'), \quad \mathring{b} = \frac{Ax}{\|A\||x|}. \end{aligned}$$

     where $w' = \kappa w$ with κ not too big.

  3. We see that downscaling may happen for the relative nominal output length $|\mathring{b}|$ in general since $\mathsf{Ker}\,(A)$ may not be trivial.
     (a) If the down scaling took place, then in terms of relative error bound, this loses significant figures.
     (b) Only if $|\mathring{b}| \simeq 10^0$ or in higher scale, loss of significant figures does not take place.

  4. Suppose A was a matrix such that $m \ge n = r = \mathsf{rank}\,(A)$, i.e., $\mathsf{Ker}\,(A)$ is trivial. In this case, singular values are arranged so that

$$\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_n > 0.$$

     The length of the nominal relative output has the lower bound

$$|\mathring{A}\mathring{x}| \ge \frac{\sigma_n}{\|A\|}|\mathring{x}| = \frac{\sigma_n}{\sigma_1}|\mathring{x}|.$$

     By re-normalizing the output,

$$s(A)s(x) = \|A\||x|\frac{\sigma_n}{\sigma_1}\Big(\frac{\sigma_1}{\sigma_n}\mathring{A}\mathring{x} + \frac{\sigma_1}{\sigma_n}\kappa w v\Big).$$

     Now, $\left|\frac{\sigma_1}{\sigma_n}\mathring{A}\mathring{x}\right| \ge 1$ and the radius $w'' = \frac{\sigma_1}{\sigma_n}\kappa w$.

     The message is, even if we know the kernel is trivial, at worst scenario the condition number is multipled by the factor $\frac{\sigma_1}{\sigma_n}\kappa$, κ not too big, and the ratio $\frac{\sigma_1}{\sigma_n}$ is the inverse of the smallest relative singular value $\frac{\sigma_1}{\sigma_n}$.

- In case one of the nominal input is zero, then the form of error is

$$s(A)^T s(x) = M_3(0 + \kappa w v_3),$$

the result is the nominal zero with no significant figure, that may represents a vector whose length $\simeq M_3 \kappa w$.

Matrix-Matrix multiplication

- Let $A \in \mathbb{R}^{\ell\times m}$, $B \in \mathbb{R}^{m\times n}$ and consider $(A,B) \mapsto AB \in \mathbb{R}^{\ell\times n}$

$$H : \mathbb{A}_1 \times \mathbb{A}_2 \to \mathbb{B}$$

- We consider the preparation of nominal A and B

$$s(A) = M_1(\mathring{A} + w_1E_1), \quad s(B) = M_2(\mathring{B} + w_2E_2), \quad |w_1|,|w_2| \le w, \quad \|E_1\|,\|E_2\| \le 1.$$

Here, $w_1, w_2 \in \mathbb{R}$ and $E_1 \in \mathbb{R}^{\ell\times m}, E_2 \in \mathbb{R}^{m\times n}$.

- For the multiplication, we find $M_3 = M_1M_2$ for the resultant typical scale appropriate.

We consider $\big(\mathbb{B}, M_3, w'\big)$ and now we seek for the bound w' below.

- We first proceed with nonzero matrix $A \in \mathbb{A}_1$ and nonzero matrix $B \in \mathbb{A}_2$.

1. We consider the preparation of nominal A and B with

$$s(A) = \|A\|(\mathring{A} + w_1E_1), \quad s(B) = \|B\|(\mathring{B} + w_2E_2), \quad \mathring{A} = \frac{A}{\|A\|}, \quad \mathring{B} = \frac{B}{\|B\|}.$$

2. Then one sample scenario via semiconductor arithmetics gives

$$\begin{aligned} s(A)s(B) &= \|A\|\|B\|\big((\mathring{A} + w_1E_1)(\mathring{B} + w_2E_2) + w_3E_3\big) \\ &= \|A\|\|B\|\big(\mathring{A}\mathring{B} + w_1E_1\mathring{B} + w_2\mathring{A}E_2 + w_1w_2E_1E_2 + w_3E_3\big) \\ &= \|A\|\|B\|\big(\mathring{A}\mathring{B} + \kappa wE_4\big) \\ &= \|A\|\|B\|(\mathring{b} + w'), \quad \mathring{b} = \frac{AB}{\|A\||B|}. \end{aligned}$$

where $w' = \kappa w$ with κ not too big.

3. We see that downscaling may happen for the relative nominal output norm $\|AB\|$ in general.
   (a) If the down scaling took place, then in terms of relative error bound, this loses significant figures.
   (b) Only if $\|\mathring{b}\| \simeq 10^0$ or in higher scale, loss of significant figures does not take place.
4. Suppose A was a matrix such that $\ell \ge m = \mathsf{rank}\,(A)$ and B was a matrix such that $m \ge n = \mathsf{rank}\,(B)$. In this case, it holds that

$$|ABx| \ge \sigma_m(A)|Bx| \ge \sigma_m(A)\sigma_n(B)|x| \quad \Longrightarrow \sigma_n(AB) \ge \sigma_m(A)\sigma_n(B)$$

For the relative matrices,

$$\sigma_n(\mathring{A}\mathring{B}) \ge \frac{\sigma_m(A)}{\sigma_1(A)}\frac{\sigma_n(B)}{\sigma_1(B)}$$

By re-normalizing the output,

$$s(A)s(x) = \|A\|\|B\|\frac{\sigma_m(A)}{\sigma_1(A)}\frac{\sigma_n(B)}{\sigma_1(B)}\Big(\frac{\sigma_1(A)}{\sigma_m(A)}\frac{\sigma_1(B)}{\sigma_n(B)}\mathring{A}\mathring{B} + \frac{\sigma_1(A)}{\sigma_m(A)}\frac{\sigma_1(B)}{\sigma_n(B)}\kappa wE_4\Big).$$

Now, $\sigma_n\Big(\frac{\sigma_1(A)}{\sigma_m(A)}\frac{\sigma_1(B)}{\sigma_n(B)}\mathring{A}\mathring{B}\Big) \ge 1$ and the radius $w'' = \frac{\sigma_1(A)}{\sigma_m(A)}\frac{\sigma_1(B)}{\sigma_n(B)}\kappa w$.

The message is, even if we know the kernel is trivial, at worst scenario the condition number is multipled by the factor $\frac{\sigma_1(A)}{\sigma_m(A)}\frac{\sigma_1(B)}{\sigma_n(B)}\kappa$, κ not too big.

- In case one of the nominal input is zero, then the form of error is

$$s(A)s(B) = M_3(0 + \kappa w E_4),$$

the result is the nominal zero with no significant figure, that may represents a matrix whose norm $\simeq M_3 \kappa w$.

Error propagation by Matrix-Matrix addition, vector-vector addition

Exercise. Similar to real number addition can be observed.

Chapter 11

QR decomposition problem analysis

1. Problem specification

The problem $\mathbf{P}_{QR}$ is to find the solver

$$H : A \mapsto (Q, R).$$

More precisely,

1. The input A is any matrix. To be concrete, let us consider

$$\mathbb{A} = \mathbb{R}^{m \times n}$$

2. The output is the pair

$$(Q, R) \in \mathbb{B} = \mathbb{B}_1 \times \mathbb{B}_2$$

where $\mathbb{B}_1$ is the set of $m \times m$ orthogonal matrices, $\mathbb{B}_2$ is the set of $m \times n$ matrices in R-form.

The second item amounts to specify objectives:

2-(a) for $k = \min\{m, n\}$

$$\begin{aligned}
\mathsf{span}\left\langle a_1 \right\rangle &\subseteq \mathsf{span}\left\langle q_1 \right\rangle \\
\mathsf{span}\left\langle a_1, a_2 \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2 \right\rangle \\
\mathsf{span}\left\langle a_1, a_2, a_3 \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2, q_3 \right\rangle \\
&\vdots \\
\mathsf{span}\left\langle a_1, a_2, a_3, \cdots, a_k \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2, q_3, \cdots, q_k \right\rangle \\
\mathsf{span}\left\langle a_1, a_2, a_3, \cdots, a_k, a_{k+1} \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2, q_3, \cdots, q_k \right\rangle \\
&\vdots \\
\mathsf{span}\left\langle a_1, a_2, a_3, \cdots, a_k, a_{k+1}, \cdots, a_n \right\rangle &\subseteq \mathsf{span}\left\langle q_1, q_2, q_3, \cdots, q_k \right\rangle.
\end{aligned}$$

2-(b) q_1, q_2, $\cdots$, q_m are orthonormal.

2. Condition of the solver H

We recall that the step is ill-conditioned:

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & w_1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tilde{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & w_1 & 0 \\ 0 & w_1 & 1 \end{pmatrix}$$

1. For A,
$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R = A.$$

2. For $\tilde{A}$,
$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad R = \text{computed accordingly.}$$

2-1. Condition of approximate solver $\tilde{H}$

We do not expect an algorithm or approximate solver $\tilde{H}$ whose condition number is small.

Having that in our mind, we go check what happens in the algorithm using reflections.

3. What happens in algorithm using reflections

The whole step $\tilde{H}$ can be broken into many steps. We consider the following j-th step solver.

1. Upstream matrix data.
   - At j-th step, A_{j-1} matrix is passed from upstream, informing also us we are to examine β-column of A_{j-1} at p-th row index.
   - If c is $\mathbb{R}^m$-column, let $c = c' + c''$,
   $$c' = (c_1, c_2, \cdots, c_{p-1}, 0, 0, \cdots, 0)^T, \quad c'' = (0, 0, \cdots, 0, c_p, c_{p+1}, \cdots, c_m)^T.$$
   Also, $\hat{c}'' = (c_p, c_{p+1}, \cdots, c_m)^T$.
   - Every prior column d of A_{j-1} are assumed: $d = d' + d''$ with $d'' = 0$.

2. $\tilde{H}_j$ step solver.
   - Its main input data is the reduced β-th column $\hat{c}''$.
   - We come up with $Q_j \in \mathbb{R}^{m\times m}$ matrix:
   $$\left(\begin{array}{c|c} \mathrm{I}' & 0 \\ \hline 0 & Q_j'' \end{array}\right)$$
   $$Q_j'' = \mathrm{I}'' - 2\hat{q}''\hat{q}''^T, \quad \hat{q}'' = \text{length normalized } \hat{\nu}''.$$
   $\hat{\nu}''$ is taken by
   $$\hat{\nu}'' = \hat{c}'' + \mathrm{copysign}(\hat{c}''[1])|\hat{c}''|\ \hat{e}_1''.$$

3. Downstream matrix data
   - We pass $A_j = Q_j A_{j-1}$ to the next solver.

The $\tilde{H}_j$ must resolve the following step objective: It takes care of the β-th column in the way that the β-th column of $A_j = Q_j A_{j-1}$ is of the form

$$(c_1, c_2, \cdots, c_{p-1}, c_p^{new}, 0, 0, 0, \cdots, 0)^T.$$

and $(j+1)$-th step solver will take care of the $(\beta+1)$-th column with $p^{new} = p + 1$.

Once step is done for $j = 1, 2, \cdots, k = \min\{m, n\}$, the resulting matrix A_k will be in R-form.

One bad case input

- Suppose it was the case that the downscaling took place on the length $|\hat{c}''|$, say its length (relative to the typical scale M) $\sim w$.
- Then, from any sample of $\hat{c}''$

$$s\big(\hat{c}''\big) = M\Big(\frac{\hat{c}''}{M} + w_1 v_1\Big) = M w_2 v_2,$$

  it produces the normal vector $\hat{\mu}''$ for the reflection with this error and then

$$\tilde{Q}'' = I'' - 2\hat{o}''\hat{o}''^T, \quad \hat{o}'' = \text{length normalized } \hat{\mu}''$$

Observations

- We can clearly see where the condition gets wrong. When $|\hat{c}''|$ is so small, the sample of it becomes spurious, the produced length normalized $\hat{\mu}''$ can be very much inaccurate.
- On the other hand, the outcome $\tilde{Q}''$ (no matter what) will be orthogonal.
- As consequences,

$$Q_j^T A_j = A_{j-1}$$

  will be also highly accurate equality.
- Despite spuriouness in $\tilde{Q}''$, $\tilde{H}_j$ will take care of the β-th column and passes the right one to the next solver:
  1. Once Q_j is multiplied to A_{j-1}, note that $\hat{c}''$ will be replaced by another one $\hat{d}''$ of very small length. What we did is to replace spurious small vector to another spurious small vector.
  2. As consequences, β-th column of A_j will be of the form

$$(c_1, c_2, \cdots, c_{p-1}, w_p^{new}, w_{p+1}^{new}, \cdots, w_m^{new})^T, \quad w_\bullet^{new} \text{ is small.}$$

     and it is certainly on the right way turning to (near) R-form, taking care of the β-th column.

Conclusions

- The eventual outcome $(\bar{Q}, \bar{R})$ is produced as a proper member of (near) $\mathbb{B}$.
- This approximate output is a true output of some sample of A by construction. The objective 2-(a) is satisfied by some nearby matrix $s(A)$.
- In other words, the algorithm is consistent.

Implementing Gram-Schmidt Orthogonalization

What is Gram-Schmidt Orthogonalization?

Let us consider family of $\mathbb{R}^m$ columns $(a_i)_{i=1}^n$. We come up with new family $(q_i)_{i=1}^n$.

Let $A \in \mathbb{R}^{m \times n}$. We recall the following for our algorithm using reflections.

- It is to find a suitable orthogonal matrix $Q \in \mathbb{R}^{m \times m}$.
- The objective is to achieve R-form, by multiplying Q from left to A:

  $$QA = R \text{ should fall into the set of } \mathbb{R}^{m \times n} \text{ matrices of R-form.}$$

- Each i-th column a_i of A is sent to Qa_i, and by this isometry the length is preserved.

We will soon observe the Gram-Schimdt Orhogonalization conducts the following:

- It is to find a suitable upper triangular matrix $R \in \mathbb{R}^{n \times n}$.
- The objective is to achieve orthonormality of columns, by multiplying R from right to A:

  $AR = Q$ should fall into the set of $\mathbb{R}^{m \times n}$ matrices, whose first $k = \min\{m, n\}$ columns are orthonormal.

- Each i-th column a_i of A is replaced by i-th column of Q,
  done by taking out some components of the column and then re-normalizing.

Taking projections out from a vector is a dangerous operation

Here, we consider an operation

$$c \mapsto \big(\mathrm{I} - qq^T\big)c$$

that is to take out from the column c the q-directional component.

- Certainly, it is the lengh-shrinking procedure, where we should worry about down scaling. When the given column is almost parallel to q already, then the result certainly lose the siginificant figures.
- Since Gram-Schmidt Orthogonalization is done by repeating such operations, we assess the quality of repeating two of them in the following two ways:
  1. $c \mapsto \big(\mathrm{I} - q_2q_2^T\big)\big(\mathrm{I} - q_1q_1^T\big)c$.
  2. $c \mapsto \big(\mathrm{I} - q_2q_2^T - q_1q_1^T\big)c$.
- Assuming q_1 and q_2 are orthogonal, they are identical mapping. In bad scenarios, two such implementations both cause problems. However, the former causes (much) less.

To take an example, we consider the orthogonalization of columns of the following matrix

$$A = \begin{pmatrix} 100 & 100 & 100 \\ 0 & 10^{-8} & 10^{-8} \\ 0 & 0 & 10^{-10} \end{pmatrix}.$$

- For a concrete calculation, we consider its typical scale $M = 100$, and the radius bound $w = 10^{-16}$.
- If calculations are made exactly, by Gram-Schmidt orthogonalization, we have

$$q_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad q_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad q_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

  Thus, $A = QR$ with $Q = \mathrm{I}$ and $R = A$.

Chapter 12

Square Matrices 2: Diagonalization of Symmetric Matrix

We are in stage: We have Elementray reflections, Elementary rotations, Elementary Projections, and QR-decomposition in our own library toolbox.

Now, we are able to build an algorithm for diagonalization of a symmetric matrix.

Backgroud

We first get acquainted with symmetric matrix.

In this chapter, $T \in \mathbb{R}^{n\times n}$. If T is symmetric,

- its eigenvalues λ_1, λ_2, $\cdots$, λ_n are all real,
- there exist the associated eigenvectors q_1, q_2, $\cdots$, q_n, and eigenvectors are orthogonal.

In short, For any T symmetric,

$$\exists(\Lambda, Q) \quad \text{such that} \quad T = Q\Lambda Q^T.$$

For the further development of perspective, let us consider an example where eigenvalues are

$$\lambda_1 = \lambda_2 = \lambda_3 = 2, \quad \lambda_4 = \lambda_5 = -2, \quad \lambda_6 = -1.$$

$$T = O \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \end{pmatrix} O^T, \quad O = \begin{pmatrix} | & | & | & | & | & | \\ o_1 & o_2 & o_3 & o_4 & o_5 & o_6 \\ | & | & | & | & | & | \end{pmatrix} \text{ is orthogonal.}$$

- We have eigenvalues 2 (repeated 3 times), -2 (repeated 2 times) and -1.
- o_1, o_2, o_3 are eigenvectors associated to 2, o_4, o_5 are to -2, and o_6 is to -1.
- We denote

  $$E_{\{2\}} = \mathsf{span}\left\langle o_1, o_2, o_3 \right\rangle, \quad E_{\{-2\}} = \mathsf{span}\left\langle o_4, o_5 \right\rangle, \quad E_{\{-1\}} = \mathsf{span}\left\langle o_6 \right\rangle.$$

- We will also use notations

  $$E_{\{2,-2\}} = \mathsf{span}\left\langle o_1, o_2, o_3, o_4, o_5 \right\rangle, \quad E_{\{2,-1\}} = \mathsf{span}\left\langle o_1, o_2, o_3, o_6 \right\rangle, \quad E_{\{2,-2,-1\}} = \mathsf{span}\left\langle o_1, o_2, o_3, o_4, o_5, o_6 \right\rangle$$

  and so on.

- Hence, in $\mathbb{R}^6$, there are mutually orthogonal subspaces $E_{\{2\}}$, $E_{\{-2\}}$, and $E_{\{-1\}}$.
  1. Every vector in $E_{\{\lambda\}}$ is an eigenvector associated to λ. If $x \in E_{\{\lambda\}}$, then $Tx = \lambda x$.
  2. Obviously, each of $E_{\{\lambda\}}$ is invariant under mapping T.

Our objective is to identify $\Sigma = \{2, -2, -1\}$, and three subspaces $E_{\{2\}}$, $E_{\{-2\}}$, and $E_{\{-1\}}$.

In terms of basis, for each of $E_{\{\lambda\}}$, picking any of its orthonormal basis (not necessarily the given ones (o_i) above) will do the job.

From now on, unless otherwise stated, eigenvalue indices are arranged so that

$$|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|.$$

If you want, one reads $|\lambda_i| = \sigma_i$, its i-th largest singular value.

Idea of algorithm

- We recall, employing the characteristic polynomial to find eigenvalues, in particular when the polynomial order gets high, is numerically bad idea.
- Instead, we make use of one of the proof of diagonalization of symmetric matrix, that is given in Analysis language, not in Algebra language.

Proposition 1. *Let $T \in \mathbb{R}^{n\times n}$ and symmetric,*

$$f : \mathbb{R}^n \to \mathbb{R}, \quad f(x) = \frac{1}{2}x^T T^T T x, \quad \mathbb{S}^{n-1} \text{ be the } (n-1)\text{-sphere } \{q \in \mathbb{R}^n \mid |q| = 1\}.$$

Let $M = \max_{q\in\mathbb{S}^{n-1}} f(q)$, q_1 be any of its maximizers, and $\lambda_1 = q_1^T T q_1$.

Then, $Tq_1 = \lambda_1 q_1$ and $|\lambda_1|$ is the largest singular value of T.

Proposition 2. *Let T, f, q_1, and λ_1 are as in Proposition 1. We consider $(n-2)$-sphere*

$$\mathbb{S}^{n-2} = \{q \in \mathbb{S}^{n-1} \mid q \perp q_1\}$$

Let $M = \max_{q\in\mathbb{S}^{n-2}} f(q)$, q_2 be any of its maximizers, and $\lambda_2 = q_2^T T q_2$.

Then, $Tq_2 = \lambda_2 q_2$ and $|\lambda_2|$ is the second largest singular value of T.

Having done this, we can take nextly

$$\mathbb{S}^{n-3} = \{q \in \mathbb{S}^{n-1} \mid q \perp q_1, \quad q \perp q_2\}$$

and so on to have Proposition 3,4, $\cdots$, n.

We will not give a proof.

- Although we have said that the global maximization of a continuous function over a compact set, in general, may not be possible to implement,
- Here, the function f is just a quadratic function, and we can handle this global maximization.
- Based on the following simple observation, we will know how to update a candidate of maximizer that does converge to a maximizer.

We get back to the above 6×6 matrix for example.

Simple observations

- Let c be any element in $\mathbb{S}^{6-1}$. Since columns of O forms a basis, c is a linear combination
$$c = t_1 o_1 + t_2 o_2 + \cdots + t_6 o_6.$$
We denoted it in blue color to emphasize that we in fact don't know any of them.
- Let $c_1 = Tc$ and consider its expansion in O basis. It must be
$$c_1 = (\lambda_1 t_1) o_1 + (\lambda_2 t_2) o_2 + \cdots + (\lambda_6 t_6) o_6.$$
  1. For each o_i-directional coefficient, the coefficient t_i gets amplification in absolute value by factor $|\lambda_i|$.
  2. More specifically in our example, from o_1 to o_5 coefficients get amplification by factor $|2| = 2$, and o_6 coefficient gets amplification by $|-1| = 1$.
  3. This may be presented in the form,
$$c_1 = |\lambda_1| \Big(\frac{\lambda_1}{|\lambda_1|} t_1 o_1 + \frac{\lambda_2}{|\lambda_1|} t_2 o_2 + \cdots + \frac{\lambda_6}{|\lambda_1|} t_6 o_1 \Big)$$
- If we multiply T once more to have $c_2 = Tc_1$,
$$c_2 = |\lambda_1|^2 \left(\left(\frac{\lambda_1}{|\lambda_1|} \right)^2 t_1 o_1 + \left(\frac{\lambda_2}{|\lambda_1|} \right)^2 t_2 o_2 + \cdots + \left(\frac{\lambda_6}{|\lambda_1|} \right)^2 t_6 o_1 \right)$$
- Hence, if we could factor out for $\alpha = 1, 2, \cdots$, $|\lambda_1|^\alpha$ from $c_\alpha = T^\alpha c$ (in fact we don't know $|\lambda_1|$), the coefficient
$$\left(\frac{\lambda_6}{|\lambda_1|} \right)^\alpha t_6 = \left(\frac{-1}{2} \right)^\alpha t_6 \quad \to \quad 0 \quad \text{as } \alpha \to \infty \text{ geometrically fast.}$$
- Even though we do not know $|\lambda_1|$, we can keep the relative ratios between coefficients, making the length of c_α normalized to be 1 always.
  1. $c_{\alpha+1}$ = length normalized Tc_α.
  2. In this case,

     Provided that it is not the case $t_1 = t_2 = \cdots = t_5 = 0$, the sixth coefficient decays to 0.
  3. To put this differentily, provided that it is not the case $t_1 = t_2 = \cdots = t_5 = 0$, we see $c_{\alpha+1}$ is strictly further tilted towards the subspace $E_{\{2,-2\}} = \mathsf{span} \langle o_1, o_2, o_3, o_4, o_5 \rangle$ than c_α is.

Based on the observations, there are only two obstacles:

1. We can make $c_{\alpha+1}$ further tilted towards $E_{\{2,-2\}}$ but provided that initially it is not the case $t_1 = t_2 = \cdots = t_5 = 0$.
2. How to drive the sequence tilted towards either of $E_{\{2\}}$ or $E_{\{-2\}}$? not towards the whole $E_{\{2,-2\}}$?

Answers

1. The problem of the possibility to have initially $t_1 = t_2 = \cdots = t_5 = 0$ is hardly worried in numerical simulations. Why?

2. To resolve the 2nd problem, we choose a suitable real number μ to work with

$$T + \mu \mathrm{I}$$

in place of T. Its eigenvalues are $\lambda_i + \mu$. There, $|2 + \mu| \neq |-2 + \mu|$.

Then the same argument shows that vectors become further tilted toward $E_{\{2+\mu\}}$ if μ was positive.

In this course, we will not further discuss the shift factor μ anymore, because on top of this main idea, the scientific library uses adapted shift choice, and a few more cost saving techniques in addition, but we will not study every detail of them.

Instead of introducing a shift choice strategy that is not used in scientific library, we will just assume from now on a suitable choice has made so that our upstream matrix T do not have above sign ambiguity.

Having two obstacles excluded, (here T is replaced by $T+\mu\mathrm{I}$) we conclude the sequence

$$\begin{aligned} c_\alpha &\to q_1 = \frac{t_1o_1+t_2o_2+t_3o_3}{|t_1o_1+t_2o_2+t_3o_3|} \in E_{\{2+\mu\}}, \quad \text{one of eigenvector, dependently on initial } c \\ c_\alpha^T T c_\alpha &\to 2+\mu. \end{aligned}$$

The second eigenvector q_2

- Having obtained q_1, we let
$$\mathbb{S}^{n-2} = \{q \in \mathbb{S}^{n-1} \mid q \perp q_1\}, \quad \text{and projection} \quad P = I - q_1 q_1^T.$$
- We do the iteration: From initial $c \in \mathbb{S}^{n-2}$
$$c_\alpha \mapsto Tc_\alpha \mapsto PTc_\alpha \mapsto \text{length normalization} = c_{\alpha+1}.$$

Convergent n sequences

Having obstacles excluded, we have in our hand for $i = 1, 2, \cdots, n$

$$\text{vector sequences} \quad \left(q_{i,\alpha}\right)_{\alpha=1}^{\infty}, \quad \text{real sequences} \quad \left(q_{i,\alpha}^T T q_{i,\alpha}\right)_{\alpha=1}^{\infty}$$

that converges to q_i and λ_i as $\alpha \to \infty$.

$$\begin{aligned} q_{i,\alpha} &\to q_i \\ q_{i,\alpha}^T T q_{i,\alpha} &\to \lambda_i. \end{aligned}$$

Simultaneous Iteration

Given the convergent finite number of sequences $\left(q_{i,\alpha}\right)_{\alpha=1}^{\infty}$, it is of course convergent uniformly in index i.

We will not give a proof but, one tempting approach to get all the limit $(q_i)_{i=1}^n$ can be arranged not in the way one after another but in the following way.

- We prepare the initial vector $q_{1,0} \in \mathbb{S}^{n-1}$.
- At the same time, we prepare the initial vector $q_{2,0}$ so that $q_{2,0} \perp q_{1,0}$.
- Further, we prepare $q_{i,0}$ for $i = 1, 2, \cdots, n$ so that $q_{i+1} \perp \mathsf{span}\left\langle q_1, q_2, \cdots, q_i \right\rangle$.
- In other words, we prepare an orthogonal matrix Q_0 consisting of columns $q_{i,0}$ for $i = 1, 2, \cdots n$.
- We let
$$C = TQ_0$$
and do the Gram-Schmidt on C. Orthogonalization of columns gives rise to the update Q_1. We will use the algorithm using reflections.

  Column-wisely, the updating is precisely what we did in previous one-after-another approach, except that the columns of Q_α have not yet reached their limits.

Summary of Simitaneous Iteration algorithm.

1. Choose initial orthogonal matrix Q_0.
2. If Q_α is given, compute
$$C = TQ_\alpha,$$
$$(Q_{\alpha+1}, R) = \mathrm{qr}_{decom}\big(C\big).$$
3. Iterate in α with respect to a certaion termination condition.

Output is the last $\bar{Q}$ and $\bar{\Lambda} = \bar{Q}^T T \bar{Q}$.

Two obstacles concerned

1. As said earlier, the first issue to pick up very special initial Q_0 is hardly worried.
2. In case the second issue with shift factor presents the obstacle, the behavior of sequence is such that Q_α oscillates between a few states instead of converging to one state, and Λ_α may be mere block diagonal. One can modify the algorithm so that indication of this behavior triggers the use of shift factor.

Simultaneous Iteration vs. QR algorithm

For diagonalization of a symmetric matrix, the algorithm bearing the name QR algorithm also is well-known. (Be careful, QR algorithm does not refer to the one achieving QR decomposition, but it refers to the diagonalization algorithm using QR decomposition we introduce below.) In exact arithmetics, it is known that Simultaneous Iteration and QR algorithm are mathematically equivalent.

The proof of mathematical equivalence is written in the textbook of Trefethen and Bau.

QR algorithm

1. We let $C_1 = T$, or $C_1 = Q_0^T T Q_0$ if we choose an initial Q_0.
2. If C_α is given, compute
$$(Q_{\alpha+1}, R) = \mathrm{qr}_{decom}\big(C_\alpha\big),$$
$$C_{\alpha+1} = Q_{\alpha+1}^T C_\alpha Q_{\alpha+1}$$
$$\Big(= RQ_{\alpha+1}\Big)$$
3. Iterate in α with respect to a certaion termination condition.

Output is the last $\bar{C} = \bar{\Lambda}$. The point of this iterations is that basis itself also is simultaneously reflected (by multiplying Q_α^T) in updating from C_α to $C_{\alpha+1}$.

To produce columns of orthogonal eigenvectors, we need to accumulate Q_α in the above codes. Initialize $\bar{Q} = Q_0$, and then in each iteration we append

- $\bar{Q} = \bar{Q}Q_{\alpha+1}$ (replacing $\bar{Q}$ by $\bar{Q}Q_{\alpha+1}$)

Chapter 13

Rectangular Matrices 3: SVD

In our library toolbox: Elementray reflections, Elementary rotations, Elementary projections, QR-decomposition, Symmetric Matrix Diagonalization.

Now we build the algorithm for SVD.

Disclaimer

Unlike others so far in our toolbox, the algorithm we are developing in this chapter is not the one mainly used in the standard scientific library. Standard algorithm used mostly is called the Divide & Conqure algorithm. It uses the proof that is of more algebraic nature and explicit formula based. It makes use of a suitable pre-treatment and then following explicit-formula based low-rank dealing techniques.

Nevertheless, we proceed with the simpler algorithm using $T = A^T A$.

Let $A \in \mathbb{R}^{m\times n}$ be any given rectangular matrix.

- We denote
$$k = \min\{m, n\}, \quad r = \mathsf{rank}\,(A), \quad (r \le k).$$
- A SVD of A is to write
$$A = O\Sigma Q^T,$$
where
  1. $O \in \mathbb{R}^{m\times m}$ is an orthogonal matrix,
  2. $Q \in \mathbb{R}^{n\times n}$ is an orthogonal matrix,
  3. $\Sigma \in \mathbb{R}^{m\times n}$ has top-left $k \times k$ diagonal block with k diagonals
$$\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0, \quad \sigma_{r+1} = \sigma_{r+2} = \cdots = \sigma_k = 0$$
and all other entries of Σ are zero.
- Let
$$T = A^T A, \quad \text{a square } n \times n \text{ matrix.}$$
T is a symmetric matrix whose eigenvalues are all real and nonnegative. Let a diagonalization of T be
$$T = Q\Lambda Q^T.$$
- The Q matrix, consists of n mutually orthonormal eigenvectors of T, is the one appeared above in SVD.
- $\sqrt{\Lambda}$ specifies Σ. More specifically, the eigenvalues must be such that
$$\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_r > 0, \quad \lambda_{r+1} = \lambda_{r+2} = \cdots = \lambda_n = 0$$
and $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, 2, \cdots, k$.
- Let $C = AQ \in \mathbb{R}^{m\times n}$. In exact arithmetics, C already consists of orthogonal columns. More specifically, C consists of n columns: from left r non-zero columns and remaining $(n - r)$ zero columns. Lengths have not been normalized.
- Length normalization of the first r-columns comprises o_j for $j = 1, 2, \cdots, r$ and supplements of o_{r+1}, o_{r+2}, $\cdots$, o_m so that the whole m-columns are orthonormal specifies the matrix O. This completes the SVD.

This pretty much explains our algorithm using $T = A^T A$.

A SVD algorithm using $T = A^T A$

Let $A \in \mathbb{R}^{m \times n}$.

1. $(\Lambda, Q) = \mathrm{sym_diag}(A^T A)$.
2. Construct a $k \times k$ block by letting $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, 2, \cdots, k$.
   $\Sigma \in \mathbb{R}^{m \times n}$ is also at hand.
3. Let $C = AQ$.
4. $(O, R) = \mathrm{qr}_{decomp}(C)$.
   In particular, we adjust this QR-decomposition so that diagonals of R are nonnegative.

The output is the (Q, Σ, O). In exact arithmetics, in fact the R in the last line, reduced up to its n-th column, must be Σ.

Home Work

Implement SVD. In addition, implement the low-rank approximation of A:

$$A_{r*} = O\Sigma_{r*}Q^T,$$

where Σ_{r*} contains only first r_* singular values and rest are cut-off.

Chapter 14

Rectangular Matrices 4: Least Square Solution

In this chapter, we consider the following problem.

- Suppose $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ are given.
- In principle, we would like to find $x \in \mathbb{R}^n$ so that

$$Ax = b.$$

- This is to impose
  1. m equations in rows
  2. for n unknowns.

Two obvious observations

1. If $b \notin \mathsf{Ran}\,(A)$, then no solution exists.
2. If $b \in \mathsf{Ran}\,(A)$ and if $\mathsf{Ker}\,(A)$ is nontrivial, then infinitely many solutions exist.

Least Square Solutions

1. In the Euclidean space $\mathbb{R}^m$, for given subspace E and for any $b \in \mathbb{R}^m$ there exists unique $b' \in E$ that is the orthogonal projection of b onto E.
   - We know howto compute b': It is done by an elementary projection matrix, using orthonormal basis of E.
   - Consider the case $E = \mathsf{Ran}\,(A)$. We know at least two algorithms to pull out orthonormal basis of $\mathsf{Ran}\,(A)$, that are pivot version of QR decomposition and SVD.
2. We let the projection b'. Since $b' \in \mathsf{Ran}\,(A)$, there exists $x_p \in \mathbb{R}^n$ so that

   $$Ax_p = b'.$$

   x_p is a particular least square solution.
3. For any $z \in \mathsf{Ker}\,(A)$, $x_p + z$ also is a solution. The set

   $$\{x_p + z \mid z \in \mathsf{Ker}\,(A)\}$$

   is called the set of all least square solutions.

Least Square by qrcp (qr with column pivoting)

For $A \in \mathbb{R}^{m\times n}$, let us consider its qrcp,

$$A = ORQ^T,$$

where

1. $O \in \mathbb{R}^{m\times m}$ is an orthogonal matrix,
2. $Q \in \mathbb{R}^{n\times n}$ is an orthogonal matrix, in particular a permutation matrix.
3. $R \in \mathbb{R}^{m\times n}$ is REF. In particular, only first r rows are nontrivial and all other rows of R are zero.

Let $k = \min\{m, n\}$ and $r = \mathsf{rank}\,(A)$. We recall,

1. $\mathsf{span}\,\langle o_1, o_2, \cdots, o_r\rangle = \mathsf{Ran}\,(A)$,
2. $\mathsf{span}\,\langle o_{r+1}, o_{r+2}, \cdots, o_m\rangle = \mathsf{Ker}\,(A^T)$,
3. For $\mathsf{Ker}\,(A)$ we have to work on it.
4. For $\mathsf{Ran}\,(A^T)$ we have to work on it.

We form $\hat{O}$ consists of first r-columns of O.

We form $\hat{R}$ consists of first r rows of R, in particular we write

$$\hat{R} = \begin{pmatrix} \hat{R}_1 & | & \hat{R}_2 \end{pmatrix}, \quad \text{with } r\times r \text{ and } r\times(n-r) \text{ blocks.}$$

Considering zero rows of R,

$$A = ORQ^T = \hat{O}\hat{R}Q^T.$$

We also form $\hat{Q}$ consists of first r-columns of Q.

Orthogonal Projection

Let $b \in \mathbb{R}^m$. Its orthogonal projection onto $\mathsf{span}\left\langle o_1, o_2, \cdots, o_r \right\rangle$ is given by

$$b' = \hat{O}\hat{O}^T b = OPO^T b, \quad \text{where } P \text{ is the very elementary projection,}$$

for example in $m = 7$, $r = 4$ case

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Particular solution of $Ax = b'$.

- Having projection b' obtained, $\exists$ a least square solution x that solves $Ax = b'$.
- Write $c = O^T b$ and $c' = Pc$ to read the equation in the way

  $$ORQ^T x = b' = OPO^T b = OPc = Oc'.$$

  Observe that

  $$x \text{ solves } ORQ^T x = Oc' \iff x \text{ solves } RQ^T x = c' \iff y = Q^T x \text{ and } y \text{ solves } Ry = c'.$$

  We also note that $c = O^T b$ is the representation of b in O-basis, $c' = Pc$ is the projection but yet in O-basis.

  We are to solve the triangular system

  $$Ry = c'$$

- In particular, observe:

  1. If the column

  $$c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_r \\ c_{r+1} \\ c_{r+2} \\ \vdots \\ c_m \end{pmatrix}, \quad c' = Pc = \begin{pmatrix} c'_1 \\ c'_2 \\ \vdots \\ c'_r \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

2. R has r rows independent, and zeros elsewhere. We are in the form of equation:

$$\left(\begin{array}{c|c} \widehat{R}_1 & \widehat{R}_2 \\ \hline 0 & 0 \end{array}\right) \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \\ y_{r+1} \\ y_{r+2} \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} c_1' \\ c_2' \\ \vdots \\ c_r' \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

- Among all such y, we choose to look for y_* of the form

$$y_* = \begin{pmatrix} y_{*1} \\ y_{*2} \\ \vdots \\ y_{*r} \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^n.$$

Thus, the essential part to obtaining such particular solution is

$$\hat{R}_1 \begin{pmatrix} y_{*1} \\ y_{*2} \\ \vdots \\ y_{*r} \end{pmatrix} = \begin{pmatrix} c_1' \\ c_2' \\ \vdots \\ c_r' \end{pmatrix}, \quad \hat{R}_1 \text{ is invertible (full rank } r\text{) upper triangular matrix.}$$

- **Backsubstitution**: It is uniquely solved from the last row using the Backsubstitution.
- We let

$$x_* = Q \begin{pmatrix} y_{*1} \\ y_{*2} \\ \vdots \\ y_{*r} \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Recap

Check that the procedure above can be reproduced by obtaining sequentially

1. $\hat{c}' = \hat{O}^T b$,
2. $\hat{R}_1 \hat{y}_* = \hat{c}', \quad x_* = \hat{Q}\hat{y}_*$.

Ker (A)

- From qrcp, more precisely from REF, $\mathsf{Ker}\,(A)$ needs to be found by extra work.
- qrcp gets the range and its orthogonal complements. To get $\mathsf{Ran}\,(A^T)$ and $\mathsf{Ker}\,(A) = \mathsf{Ran}\,(A^T)^\perp$, of course, doing qrcp on A^T will do the job.

  If A is row equivalent to A', then A^T is column equivalent to A'^T. Thus, if REF of A is already obtained, then we do not need to do the job from the first page, but work with the REF of A.
- One check point: R is the REF of AQ, i.e., $R = O^T(AQ)$. Thus,

$$\mathsf{Ker}\,(A) = \big\{Qz \mid z \in \mathsf{Ker}\,R\big\} = Q\mathsf{Ker}\,(R).$$

Ker (R)

- More precisely, we gets $\mathsf{Ran}\,(R^T)$ and $\mathsf{Ran}\,(R^T)^\perp\Big(= \mathsf{Ker}\,(R)\Big)$.
- Consider R^T, consisting of first r columns and followed zero columns.
- For QR-decomposition of R^T, zero columns are not effective. We proceed with QR-decomposition of $\hat{R}^T \in \mathbb{R}^{n\times r}$ of full rank:

  $\hat{R}^T = \bar{Q}\bar{R}$, $\quad \bar{Q} \in \mathbb{R}^{n\times n}$ orthogonal, $\bar{R} \in \mathbb{R}^{n\times r}$ in full rank R-form. (REF as it is)

  $\bar{R}$ consists of $r \times r$ top upper triangular block, and zeros elsewhere.

  We extract

$$\mathsf{Ran}\,(\hat{R}^T) = \mathsf{span}\,\big\langle \bar{q}_1, \bar{q}_2, \cdots, \bar{q}_r\big\rangle, \quad \mathsf{Ran}\,(\hat{R}^T)^\perp = \mathsf{span}\,\big\langle \bar{q}_{r+1}, \bar{q}_{r+2}, \cdots, \bar{q}_n\big\rangle.$$

Set of least square solutions

The set

$$\big\{x_* + Qz \mid z \in \mathsf{span}\,\big\langle \bar{q}_{r+1}, \bar{q}_{r+2}, \cdots, \bar{q}_n\big\rangle\big\}$$

contains all least square solutions.

Remark 14.1. To see what has happened, substitute $\hat{R}$ by $\bar{R}^T\bar{Q}^T$ to see

$$A = O\left(\begin{array}{c|c} \bar{R}_1^T & 0 \\ \hline 0 & 0 \end{array}\right)\bar{Q}^TQ^T, \quad \bar{R}_1^T \text{ is full rank lower triangular.}$$

This also bears the name that is the complete QR decomposition.

One may obtain a particular solution after achieving the complete QR. The obtained particular solution in this case will have additional property, orthogonal to $\mathsf{Ker}\,(A)$. One obtained in the above does not have this property in general.

Summary of algorithm

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ is given.

1. $(O, R, Q) = qrcp(A)$.
   Identifying numerically $r = \mathsf{rank}\,(A)$, form also $\hat{O}$, $\hat{R}$ accordingly.
2. $\hat{y}_* = \text{backsubs}(\hat{R}_1, \hat{c}')$, where $\hat{c}' = \hat{O}^T b$.
3. $x_* = \hat{Q}\hat{y}_*$.
4. $(\bar{Q}, \bar{R}) = qr(\hat{R}^T)$.
5. Any $x_* + Qz$ with $z \in \mathsf{span}\left\langle \bar{q}_{r+1}, \bar{q}_{r+2}, \cdots, \bar{q}_n \right\rangle$ is a least square solution.

Remark 14.2. If there are more equations than unknowns, i.e., $m \geq n$, and if one is certain that it is the case of the full rank, then one simply uses the simplest QR decomposition. This will pick up the unique least square solution, and no step to compute $\mathsf{Ker}\,(A)$.

Least Square by svd

Having seen the Least Square by qrcp, the one by svd is simply explained by the final part.

We let a svd of A is given

$$A = O\Sigma Q^T = \hat{O}\hat{\Sigma}\hat{Q}^T.$$

Obtain sequentially

1. $\hat{c}' = \hat{O}^T b$,
2. $\hat{\Sigma}\hat{y}_* = \hat{c}', \quad x_* = \hat{Q}\hat{y}_*$.
3. $\left\{x_* + z \mid z \in \mathsf{span}\left\langle q_{r+1}, q_{r+2}, \cdots, q_n \right\rangle\right\}$.

In particular, in the second step, we simply solve a diagonal system.

Summary of algorithm

Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ is given.

1. $(O, \Sigma, Q) = svd(A)$.
   Identifying numerically $r = \mathsf{rank}\,(A)$, form also $\hat{O}$, $\hat{\Sigma}$, and $\hat{Q}$ accordingly.
2. $\hat{y}_* = \text{diagonal-invert}(\hat{\Sigma}, \hat{c}')$, where $\hat{c}' = \hat{O}^T b$.
3. $x_* = \hat{Q}\hat{y}_*$.
4. Any $x_* + z$ with $z \in \mathsf{span}\left\langle q_{r+1}, q_{r+2}, \cdots, q_n \right\rangle$ is a least square solution.

Chapter 15

Square Matrices 3: Solving Symmetric Positive Definite system

The other extreme to the previous fully general case is to solve a system

$$Mx = b, \quad M \in \mathbb{R}^{n\times n} \text{ is symmetric positive definite.}$$

This is one of the nicesest form of the equations.

$M \in \mathbb{R}^{n\times n}$ is symmetric positive definite if M is symmetric and

$$\text{for all } x \neq 0, \quad x^T M x > 0.$$

Equivalently, M is symmetric and eigenvalues are such that

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n > 0, \quad \text{and} \quad x^T M x \geq \lambda_n |x|^2.$$

In this chapter, we will introduce two algorithms:

1. Conjugate Gradient
2. A dedicated LU-decomposition $M = L^T L$, that is called the Cholesky decomposition.

Perspective on spd system

- For a given spd system, we can immediately think of at least two algorithms:
  1. Diagonalize $M = Q\Lambda Q^T$, and do similar what we did before.
  2. QR-decomposition of $M = OR$, and do similar what we did before.
- QR-decomposition, if you counts number of arithmetic operations, operation counts are of $O(n^3)$. To compete, any algorithm of $O(n^{3+\alpha})$ will not be seriously considered.
- For Diagonalization of symmetric matrix and SVD, our implementations were a little careless on cost-saving, and implementations as they are in this note certainly exceeds $O(n^3)$,

  however, it is known that both SVD and Diagonalization of symmetric matrix can be implemented to be $O(n^3)$.
- Two more algorithms, Conjugate Gradient, and the special LU decompostions are also $O(n^3)$.
- How the two algorithms achieve $O(n^3)$ is the key things to watch, and we will take extra care not to miss on this matter when we write codes, particularly in this chapter.

Conjugate Gradient

- Conjugate Gradient is a beautiful algorithm combining so many brilliant ideas.
- We take one particular way of interpreting it below.

Stage 1

- Suppose we are given vectors $p_1, p_2, \cdots, p_n$ such that
  1. They are linearly independent, forming the basis.
  2. In particular, they satisfy the property

$$\text{If } i \neq j \quad p_i^T M p_j = 0.$$

  We are not speaking of how to, or even whether it is possible or not.
- Since (p_i) forms a basis, the solution x will be a linear combination

$$x = c_1 p_1 + c_2 p_2 + \cdots + c_{n-1} p_{n-1} + c_n p_n,$$

  c_i are in blue color to emphasize that they are unknowns.
- If so, by multiplying M, we have

$$Mx = b = c_1 M p_1 + c_2 M p_2 + \cdots + c_{n-1} M p_{n-1} + c_n M p_n.$$

- Then coefficients c_i is easily obtained from:

$$\text{For } i = 1, 2, \cdots n, \quad p_i^T b = c_i p_i^T M p_i, \quad \text{in particular } p_i^T M p_i > 0.$$

Now, we examine the two questions we left behind:

1. Is it possible to have such a basis?
2. How do we make them ?

The answer to these two questions are affirmatively given, again, by Gram-Schmidt.

To be more systematic, we make use of the fact that M induces an inner product:

$$\langle x, y \rangle_M := x^T M y.$$

$\langle \cdot, \cdot \rangle$ is called an inner product if it is linear in each variable, symmetric, and

$$\forall x \quad \langle x, x \rangle \geq 0 \quad \text{and} \quad \langle x, x \rangle = 0 \Longleftrightarrow x = 0.$$

It is the inner product that gives the concept of orthogonality. To dicern the new concept from the one $x^T y = 0$, we will say x and y are M-orthogonal if

$$\langle x, y \rangle_M = x^T M y = 0.$$

What requires in the above is to come up with basis that are mutually M-orthogonal.

Stage 2 : M-Gram-Schmidt

- If $a_1,\ a_2,\ \cdots,\ a_n$ are any linearly independent vectors, one can perform M-Gram-Schmidt.
- We begin with
$$p_1 = a_1.$$
- Nextly,
$$\begin{aligned} p_2 &= a_2 - \Big(\frac{p_1^T M a_2}{p_1^T M p_1}\Big) p_1 \\ &= \Big(I - \frac{p_1 (Mp_1)^T}{p_1^T (Mp_1)}\Big) a_2. \end{aligned}$$
Since $a_2 \notin \mathsf{span}\,\langle a_1 \rangle = \mathsf{span}\,\langle p_1 \rangle$, p_2 must be nonzero.
- Nextly,
$$\begin{aligned} p_3 &= a_3 - \Big(\frac{p_1^T M a_3}{p_1^T M p_1}\Big) p_1 - \Big(\frac{p_2^T M a_3}{p_2^T M p_2}\Big) p_2 \\ &= \Big(I - \frac{p_2 (Mp_2)^T}{p_2^T (Mp_2)}\Big)\Big(I - \frac{p_1 (Mp_1)^T}{p_1^T (Mp_1)}\Big) a_3. \end{aligned}$$
Again, since $a_3 \notin \mathsf{span}\,\langle a_1, a_2 \rangle = \mathsf{span}\,\langle p_1, p_2 \rangle$, p_3 must be nonzero.
- We do this until we obtain p_n.

As announced, we check the operation counts.

- Matrix-vector multiplication is $O(n^2)$
- $x^T y$ is $O(n)$.
- Provided that Mp_i are once computed and re-used ever after, at p_k-obtaining step, operation counts will be
$$O(n^2 + nk).$$
If we are careless about re-using Mp_k, it falls down to $O(n^2 + n^2 k)$.
- Total operation counts are
$$\sum_{k=1}^{n} O(n^2 + nk) = O(n^3).$$

This M-Gram-Schmidt managed to be an $O(n^3)$ algorithm. Thus, it is a decent Solver.

Even nicer thing is that there are still something we can exploit: the initial $a_1, a_2, \cdots, a_n$ are arbiturary as long as they are independent. There should be the "clever" initial.

Stage 3

- We will decide a_{k+1}, importantly, after seeing what has been $p_1, p_2, \cdots, p_k$.
- Picking a_{k+1} that are M-orthogonal to $p_1, p_2, \cdots, p_k$ at once would be too much.
- But, it turns out that there is a way to pick a_{k+1} that are already M-orthogonal to $p_1, p_2, \cdots, p_{k-1}$. This says that for its M-Gram-Schmidt, there will be only one projection taking out.

This beautifully combines the geometric idea with respect to Gradient Descent. Let us look at the method with repect to the problem $Mx = b$.

1. Define
$$\varphi(x) = \frac{1}{2}x^T Mx - b^T x, \quad \varphi : \mathbb{R}^n \to \mathbb{R}.$$
It is a constant shift of a same kind function
$$\frac{1}{2}(x - x_*)^T M(x - x_*) = \frac{1}{2}x^T Mx - b^T x + \frac{1}{2}{x_*}^T Mx_*,$$
where x_* is a solution of $Mx = b$.
2. That M is spd means M is a dialation with positive factors in eigenvector directions:
   (a) The contour set $\{x \mid \varphi(x) = t\}$ for $t \in \mathbb{R}$ will be $(n-1)$-dimensional hyperellipse,
   (b) and x_* is the center of contour sets, as the unique minimizer of φ.
3. Hence, Gradient Descent, approximate solution is descending toward the negative gradient direction, comes as a natural algorithm.
   However, φ is not just a function but is a quadratic polynomial, and we can certainly do better for its minimization ! : Conjugate Gradient Descent.
   Let us take the simple 2 case by picture:

4. Let us also prepare the formula of the negative gradient of φ at a given point x_0:
$$-\nabla\varphi(x_0) = b - Mx_0.$$

Stage 3: Geometric explanation

We describe below what Conjugate Gradient does geometrically. It is a Descending algorithm.

For simplicity, we begin at the origin $x_0 = 0$.

1. The first p_1 vector, the direction we descend, is chosen freely of M-orthogonality. At $x_0 = 0$, a good choice for the first p vector is the negative gradient direction

$$a_1 = p_1 = -\nabla\varphi(x_0) = b.$$

   How far do we take a step in the p_1 direction ? In p_1 direction from x_0, φ at first starts decreasing. You will stop where φ ceases to decrease, and there, the direction p_1 is tangential to the contour set.

   The point is the x_1, perhaps closer to x_* than x_0 is, since $\varphi(x_1) \leq \varphi(x_0)$.

2. At x_1, we know

   (a) p_1 is tangential to contour set at x_1.

   (b) $-\nabla\varphi(x_1)$ that is $b - Mx_1$ is normal to contour set at x_1, unless $b - Mx_1 = 0$.

   (c) We choose to pick

$$a_2 = -\nabla\varphi(x_1) = b - Mx_1.$$

   Now we choose to pick the second vector p_2

   (a) M-orthogonal to p_1. (Many choices since $\langle p_1 \rangle^{\perp_M}$ is $(n-1)$ dimensional.)

   (b) Consider the two dimensional plane $\mathsf{span}\langle p_1, a_2 \rangle$. We choose p_2 on this plane.

   Again, how far do we take a step in the new direction p_2 from x_1 ? Until φ ceases to decrease. That is the point x_2.

3. At x_2, we know

   (a) p_2 is tangential to contour set at x_2.

   (b) $-\nabla\varphi(x_2)$ that is $b - Mx_2$ is normal to contour set at x_2, unless $b - Mx_2 = 0$.

   (c) We choose to pick

$$a_3 = -\nabla\varphi(x_2) = b - Mx_2.$$

   Now we choose to pick the second vector p_3

   (a) M-orthogonal to $\mathsf{span}\langle p_1, p_2 \rangle$. (Many choices since $\langle p_1, p_2 \rangle^{\perp_M}$ is $(n-2)$ dimensional.)

   (b) Consider the three dimensional subspace $\mathsf{span}\langle p_1, p_2, a_3 \rangle$. We choose p_3 on this subspace.

   Again, how far do we take a step in the new direction p_3 from x_2 ? Until φ ceases to decrease. That is the point x_3.

4. We will prove that the n-th update x_n must be the solution, unless solution is found at k-th update $k \leq n$.

Stage 3: Algebraic explanation

We review the procedure calculating concretely.

For simplicity, we begin at the origin $x_0 = 0$.

1. At $x_0 = 0$, we have chosen the negative gradient

$$a_1 = p_1 = -\nabla\varphi(x_0) = b.$$

The point

$$x_1 = x_0 + c_1 p_1.$$

c_1 can be computed from two reasonings.

(a) Provisionally, we will eventually have

$$\begin{aligned} x_* &= c_1 p_1 + c_2 p_2 + c_3 p_3 + \cdots + c_n p_n, \quad (p_i) \text{ are mutually } M\text{-orthogonal} \\ \text{(we will write} \quad x_k &= 0 + c_1 p_1 + c_2 p_2 + c_3 p_3 + \cdots + c_k p_k, \quad k = 0, 1, 2, \cdots, n) \\ Mx_* = b &= c_1 Mp_1 + c_2 Mp_2 + c_3 Mp_3 + \cdots + c_n Mp_n, \end{aligned}$$

and thus $c_1 = \frac{p_1^T b}{p_1^T M p_1}$.

(b) Or, we monitor how $f(t) = \varphi(0 + tp_1) = \frac{t^2}{2} p_1^T M p_1 - t p_1^T b$ dereases in $t > 0$. It will stop at t_* such that

$$f'(t_*) = 0 = t_* p_1^T M p_1 - p_1^T b \quad \Longleftrightarrow \quad t_* = \frac{p_1^T b}{p_1^T M p_1}.$$

In conclusion, the first update

$$x_1 = x_0 + c_1 p_1, \quad c_1 = \frac{p_1^T b}{p_1^T M p_1}.$$

2. At x_1, we let

$$a_2 = -\nabla\varphi(x_1) = b - Mx_1 = b - c_1 Mp_1.$$

Assuming $a_2 \neq 0$, we pick $p_2 \in \mathsf{span}\left\langle p_1, a_2 \right\rangle$ by doing M-Gram-Schmidt of a_2:

(a) First of all, $a_2 \notin \mathsf{span}\left\langle p_1 \right\rangle$. Recall c_1 is precisely so that $p_1^T a_2 = 0$,

$$a_2 \in \mathsf{span}\left\langle p_1 \right\rangle \cap \mathsf{span}\left\langle p_1 \right\rangle^{\perp} \quad \text{would imply} \quad a_2 = 0.$$

(b) Let $p_2 = a_2 - \beta p_1, \quad \beta = \frac{p_1^T M a_2}{p_1^T M p_1}$.

Having obtained the new direction p_2, we are to take a step towards x_2. Before we take a step, we record what are true at this moment.

(a) p_1 and p_2 are M-orthogonal.

(b) $a_1 = p_1$ and a_2 are orthogonal upon checking $p_1^T a_2 = 0$.

(c) $\mathsf{span}\left\langle a_1, a_2 \right\rangle = \mathsf{span}\left\langle p_1, p_2 \right\rangle$.

(d) $Mp_1 \in \mathsf{span}\left\langle p_1, a_2 \right\rangle = \mathsf{span}\left\langle p_1, p_2 \right\rangle$.

Now, we take a step in the new direction p_2,

$$x_2 = x_1 + c_2 p_2 = x_0 + c_1 p_1 + c_2 p_2.$$

Let us once more monitor $g(t) = \varphi(x_1 + tp_2) = \frac{t^2}{2} p_2^T M p_2 - t p_2^T b + \text{constant terms}$.

$$g'(t_*) = 0 = t_* p_2^T M p_2 - p_2^T b \quad \Longleftrightarrow \quad t_* = \frac{p_2^T b}{p_2^T M p_2}.$$

In conclusion, the second update

$$x_2 = x_1 + c_2 p_2 = x_0 + c_1 p_1 + c_2 p_2, \quad c_2 = \frac{p_2^T b}{p_2^T M p_2}.$$

3. At x_2, we let
$$a_3 = -\nabla\varphi(x_2) = b - Mx_2 = b - c_1 M p_1 - c_2 M p_2.$$
Assming $a_3 \neq 0$, we pick $p_3 \in \mathsf{span}\left\langle p_1, p_2, a_3 \right\rangle$ by doing M-Gram-Schmidt of a_3:

   (a) First of all, $a_3 \notin \mathsf{span}\left\langle p_1, p_2 \right\rangle$. Recall c_1 and c_2 are precisely so that $p_1^T a_3 = 0$ and $p_2^T a_3 = 0$,
   $$a_3 \in \mathsf{span}\left\langle p_1, p_2 \right\rangle \cap \mathsf{span}\left\langle p_1, p_2 \right\rangle^{\perp} \quad \text{would imply} \quad a_3 = 0.$$

   (b) We check the important property that, a_3 is already M-orthogonal to p_1. This is because, we have just observed
   $$a_3 \perp \mathsf{span}\left\langle p_1, p_2 \right\rangle \quad \text{but we did check } Mp_1 \in \mathsf{span}\left\langle p_1, p_2 \right\rangle.$$

   (c) We only need to make one operation:
   $$p_3 = a_3 - \beta p_2, \quad \beta = \frac{p_2^T M a_3}{p_2^T M p_2}.$$

Before we make a step towards x_3, we record what are true at this moment.

   (a) p_1, p_2, p_3 are M-orthogonal.
   (b) a_1, a_2, a_3 are orthogonal.
   (c) $\mathsf{span}\left\langle a_1, a_2, a_3 \right\rangle = \mathsf{span}\left\langle p_1, p_2, p_3 \right\rangle$.
   (d) $Mp_2 \in \mathsf{span}\left\langle p_1, Mp_1, a_3 \right\rangle \subset \mathsf{span}\left\langle p_1, p_2, a_3 \right\rangle = \mathsf{span}\left\langle p_1, p_2, p_3 \right\rangle$.

Now, the third update

$$x_3 = x_2 + c_3 p_3 = x_0 + c_1 p_1 + c_2 p_2 + c_3 p_3, \quad c_3 = \frac{p_3^T b}{p_3^T M p_3}.$$

4. Let us do the one more iteration at x_3, we let

$$a_4 = -\nabla\varphi(x_3) = b - Mx_3 = b - c_1Mp_1 - c_2Mp_2 - c_3Mp_3.$$

Assuming $a_4 \neq 0$, we pick $p_4 \in \mathsf{span}\left\langle p_1, p_2, p_3, a_4\right\rangle$ by doing M-Gram-Schmidt of a_4:

(a) First of all, $a_4 \notin \mathsf{span}\left\langle p_1, p_2, p_3\right\rangle$. Recall c_1, c_2, and c_3 are precisely so that $p_1^T a_4 = 0$, $p_2^T a_4 = 0$, and $p_3^T a_4 = 0$,

$$a_4 \in \mathsf{span}\left\langle p_1, p_2, p_3\right\rangle \cap \mathsf{span}\left\langle p_1, p_2, p_3\right\rangle^{\perp} \quad \text{would imply} \quad a_4 = 0.$$

(b) We check the important property that, a_4 is already M-orthogonal to p_1 and p_2. This is because, we have just observed

$$a_4 \perp \mathsf{span}\left\langle p_1, p_2, p_3\right\rangle \quad \text{but we did check } Mp_1, Mp_2 \in \mathsf{span}\left\langle p_1, p_2, p_3\right\rangle.$$

(c) We only need to make one operation:

$$p_4 = a_4 - \beta p_3, \quad \beta = \frac{p_3^T M a_4}{p_3^T M p_3}.$$

Before we make a step towards x_4, we record what are true at this moment.

(a) p_1, p_2, p_3, p_4 are M-orthogonal.
(b) a_1, a_2, a_3, a_4 are orthogonal.
(c) $\mathsf{span}\left\langle a_1, a_2, a_3, a_4\right\rangle = \mathsf{span}\left\langle p_1, p_2, p_3, p_4\right\rangle$.
(d) $Mp_3 \in \mathsf{span}\left\langle p_1, Mp_1, Mp_2, a_4\right\rangle \subset \mathsf{span}\left\langle p_1, p_2, p_3, a_4\right\rangle = \mathsf{span}\left\langle p_1, p_2, p_3, p_4\right\rangle$.

Now, the 4-th update

$$x_4 = x_3 + c_4 p_4 = x_0 + c_1 p_1 + c_2 p_2 + c_3 p_3 + c_4 p_4, \quad c_4 = \frac{p_4^T b}{p_4^T M p_4}.$$

One can visualize this will end up with having

$$\begin{aligned} x_n &= c_1 p_1 + c_2 p_2 + c_3 p_3 + \cdots + c_n p_n, \\ a_{n+1} &= b - c_1 M p_1 - c_2 M p_2 - \cdots - c_n M p_n = 0. \end{aligned}$$

Stage 3: Pseudo code

Since practically we may want to start from a good guess x_0, we consider a new problem

$$M(x - x_0) = b - Mx_0, \quad \text{or} \quad M\bar{x} = \bar{b}, \quad \bar{x} = x - x_0, \quad \bar{b} = b - Mx_0$$

to look for solution $x = x_0 + \bar{x}$.

1. Initialize x_0, and $\bar{b} = b - Mx_0$.
2. Let $a_1 = p_1 = \bar{b}$.
3. If $|a_1| \simeq 0$, exit with current x_0. Otherwise, do the iteration in $k = 1, 2, \cdots, n$:

   (a) $(Mp_k) = Mp_k$. Here we have $O(n^2)$ operation counts.
   (b) $c_k = \frac{p_k^T \bar{b}}{p_k^T (Mp_k)}$
   (c) $x_k = x_{k-1} + c_k p_k$.
   (d) $a_{k+1} = a_k - c_k(Mp_k)$. If $|a_{k+1}| \simeq 0$, exit with current x_k. Otherwise,
   (e) $\beta_{k+1} = \frac{(Mp_k)^T a_{k+1}}{p_k^T (Mp_k)}$
   (f) $p_{k+1} = a_{k+1} - \beta_{k+1} p_k$.

One another equivalent formulas are

$$c_k = \frac{p_k^T \bar{b}}{p_k^T (Mp_k)} = \frac{p_k^T \left(a_k + \sum_{i=1}^{k-1} c_i Mp_i\right)}{p_k^T (Mp_k)} = \frac{p_k^T a_k}{p_k^T (Mp_k)} = \frac{(a_k - \beta_k p_{k-1})^T a_k}{p_k^T (Mp_k)} = \frac{a_k^T a_k}{p_k^T (Mp_k)},$$

$$\beta_{k+1} = \frac{(Mp_k)^T a_{k+1}}{p_k^T (Mp_k)} = \frac{((a_{k+1} - a_k)/(-c_k))^T a_{k+1}}{((a_{k+1} - a_k)/(-c_k))^T p_k}$$

$$= \frac{(a_{k+1} - a_k)^T a_{k+1}}{(a_{k+1} - a_k)^T p_k} = \frac{a_{k+1}^T a_{k+1}}{-a_k^T p_k} = \frac{a_{k+1}^T a_{k+1}}{-a_k^T a_k} \quad \text{using orthogonalites.}$$

Re-written iteration:

   (a) $(Mp_k) = Mp_k$. Here we have $O(n^2)$ operation counts.
   (b) $c_k = \frac{a_k^T a_k}{p_k^T (Mp_k)}$
   (c) $x_k = x_{k-1} + c_k p_k$.
   (d) $a_{k+1} = a_k - c_k(Mp_k)$. If $|a_{k+1}| \simeq 0$, exit with current x_k. Otherwise,
   (e) $\beta_{k+1} = \frac{a_{k+1}^T a_{k+1}}{a_k^T a_k}$.
   (f) $p_{k+1} = a_{k+1} + \beta_{k+1} p_k$.

Other than computations in red color, one re-uses the pre-existing values.

Remark on being $O(n^3)$ and others

- Although it cannot beat $O(n^3)$ in general, one notices that if convergence to x_* is sufficiently nice with iterations fewer than n, say ℓ times, operational counts will be

$$O(\ell n^2).$$

- If M happens to be sparse, say tri-diagonal, then actually (Mp_k) costs $O(n)$. Then this will work as

$$O(\ell n) \leq O(n^2).$$

  For some many important matrices arise in PDE, Conjugate Gradient sometimes extraordinarily fast.

- Since it does not actually invert M but produces updates depending on rhs b in iterations, for situation we are supposed to solve

$$Mx = b_1, \quad Mx = b_2, \quad Mx = b_3, \quad \cdots$$

  we better invert M indeed, say by diagonalization or by other algorithms.